

Part II — First principles: how money moves around an engine

Descriptive reference. Sources are cited inline and listed at the end. Takes no view.

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PART II

First principles: how money moves around an engine

This Part explains how an aircraft or an engine turns into cash flows: the lease that rents it out, the reserves that pay for the life a lessee uses up, the settlement at the end of the lease, the rent an engine earns on its own, the trade in engines with a little life left, and the debt that funds the owner. A reader who finishes it will be able to read a lease summary, a lessor's income statement and a securitization term sheet with every term understood.

WHERE THIS PART SITS IN THE BUILD

Part I described the CFM56 (the engine family Part I defined) as a machine: how it comes apart, which parts are scrapped on a clock, what a shop visit costs, and what the engine is worth at any point in its life. This Part takes that object and follows the money around it. Almost every dollar an engine earns for its owner passes through a contract, usually an operating lease, and almost every dollar its owner borrows is secured on that contract. So the lease is the unit of analysis here, the way the module was the unit in Part I. Part III, which follows, supplies the context: the fleet, the retirements that did not happen, the new engines that arrived late, and a translation table that turns each fact into a requirement on the aftermarket. Nothing in this Part depends on anything written after it. Parts VIII and XI, which cover FTAI's leasing segment, its third-party capital vehicles and its peers, depend on every definition here.

12. The operating lease, term by term

An aircraft or engine earns money for its owner in three ways. It earns rent while it is leased. It earns payments for the maintenance life a lessee consumes while flying it. And it earns a price when it is sold, either whole or in parts. The first two flows are written into a lease, and the third is shaped by the condition the lease requires the asset to come back in. This section takes the lease apart term by term. Section 13 works the maintenance terms through a full example, section 14 applies the whole to an engine on its own, and sections 15 to 17 follow the asset to the end of its life and to the lenders who fund it.

The two parties have plain names. The lessor is the owner who rents the asset out. The lessee is the operator, almost always an airline, who rents it. The research found no reason to make either word more technical than that. [definition assembled in the research, D6 §2.1]

OPERATING LEASE

An operating lease is a contract under which the owner (the lessor) rents an aircraft or engine to an operator (the lessee) for a fixed term in exchange for periodic rent, while the lessor keeps title to the asset and keeps the risk that it is worth more or less than expected when it comes back. Nearly all of FTAI's leases, and nearly all of Willis Lease Finance's, are operating leases. The alternative is the finance lease, defined next. [FTAI 10-K FY2025; Willis Lease Finance Q3 2025 release; definition assembled in the research, D6 §2.1]

FINANCE LEASE (SALES-TYPE LEASE)

A finance lease transfers substantially all the economic risks and rewards of owning the asset to the lessee, so that in substance it is a loan secured on the asset rather than a rental. Willis Lease Finance Corporation (WLFC) reports a small book of "investments in sales-type leases", \$16.3M of its \$2,888.5M lease portfolio at 30 September 2025, which are finance leases by another name; FTAI reports none. [WLFC Q3 2025 release; definition assembled in the research, D6 §2.1]

The distinction matters for everything that follows. Under an operating lease the lessor still owns the engine when it comes back, so the lessor cares what condition it comes back in, which is why the maintenance terms in §13 exist. Under a finance lease the lessee effectively owns it and the lessor is a lender. This Part is about operating leases unless it says otherwise.

12.1 TERM, RENT AND THE LEASE RATE FACTOR

The term is the fixed period of the lease. For a mid-life narrowbody it typically runs several years; FTAI reported a weighted-average remaining term of 44 months on its aircraft on lease and 38 months on its engines on lease at 31 December 2025. [FTAI 10-K FY2025] Those are remaining terms, not the terms the leases were signed at; the research did not retrieve new-lease terms at signing or the mix of short against long engine leases in FTAI's book (D6 U15). Engine leases divide into short-term and long-term, which Part I §6.4 defined; §14 carries them.

LEASE RATE (RENT)

The lease rate is the periodic rent, quoted per month. The sourced points in this primer are about \$228,000 a month for a 12-year-old Boeing 737-800 and about \$220,000 for a comparable Airbus A320-200 as of July 2025, and about \$100,000 a month for a CFM56-7B spare engine in 2024, all from IBA (an appraiser), against \$42,000 to \$48,000 a month for a CFM56-7B in 2026 from a consultancy blog that may be describing a green-time engine. The two engine figures are carried as a pair throughout this primer and never averaged. [IBA, Jul 2025, via India Seatrade News, 6 Sep 2026; IBA, Apr 2024, via AviTrader; Safe Fly Aviation, 2026 (consultancy blog); reconciliation R-028]

Rent on its own says little about whether a lease is a good one for the owner, because a \$228,000 rent on a \$15M aircraft is a very different thing from the same rent on a \$30M aircraft. The industry therefore quotes rent as a percentage of what the asset is worth.

LEASE RATE FACTOR (LRF)

The lease rate factor is the monthly rent divided by the asset's current market value, expressed as a percentage. An LRF of 1.0% means annual rent is 12% of value. It is the leasing industry's shorthand for gross yield, before depreciation, interest and maintenance. It rises with asset age, because older assets carry more residual and maintenance risk and lose value faster in percentage terms, and it rises with interest rates. No source the research reached states a lease rate factor for FTAI, for Willis Lease or for the CFM56 as a type; every figure in this primer is derived, and §14 shows each with its basis. [definition assembled in the research, D6 §2.1, D11 §2.2; reconciliation R-030]

WORKED EXAMPLE 2.1 — READING A LEASE RATE FACTOR

Inputs: rent of \$228,000 a month for a 12-year-old 737-800 (sourced: IBA, July 2025). The aircraft's half-life market value was not retrieved by the research (D6 U9), so the value is the unknown in this example and the lease rate factor is the variable.

$$\begin{aligned} \text{LRF} &= \text{rent} \div \text{value} \\ \text{at LRF 0.8\% a month: value} &= 228,000 \div 0.008 = \$28.5\text{M} \\ \text{at LRF 1.0\% a month: value} &= 228,000 \div 0.010 = \$22.8\text{M} \\ \text{at LRF 1.2\% a month: value} &= 228,000 \div 0.012 = \$19.0\text{M} \\ \text{annual gross yield} &= \text{LRF} \times 12: 9.6\%, 12.0\%, 14.4\% \end{aligned}$$

So the same rent is a 9.6% gross yield on a \$28.5M aircraft and a 14.4% gross yield on a \$19.0M one. The identity cuts both ways: a lessor who knows the value quotes the lease rate factor, and a buyer who knows the rent and the factor other lessors accept can back out what the aircraft is worth to them. Limits of the example: the three factors are chosen to show the arithmetic and are not market data; the aircraft value was not sourced; and the factor for a specific lease depends on the lessee's credit, the term and the return conditions below. [IBA, Jul 2025; D6 U9]

12.2 THE SECURITY DEPOSIT**SECURITY DEPOSIT**

A security deposit is cash, or a bank letter of credit, that the lessee posts at signing. The lessor holds it against default and returns it at redelivery if the lessee has performed. It is a credit instrument, not a maintenance instrument: it protects the lessor against unpaid rent, not against life consumed from the engines. FTAI's balance sheet carries "security deposits" as a liability; the amounts by year were not returned by the research (D6 U7). [FTAI 10-K FY2025; definition assembled in the research, D6 §2.1]

The deposit is the first of three pots of the lessee's money that a lessor may hold. The other two, maintenance reserves and end-of-lease compensation, are maintenance instruments and are the subject of §13. They are defined here because the lease contains them.

12.3 MAINTENANCE RESERVES AND END-OF-LEASE COMPENSATION

Every hour an engine flies uses up some of the life that was in it when the lease began: the exhaust-gas temperature margin that a performance restoration will have to recover, and the cycles on each life-limited part (Part I §4 and §5). A lease must say who pays for that. There are two regimes.

MAINTENANCE RESERVES (SUPPLEMENTAL RENT; MAINTENANCE RENTS)

Maintenance reserves are periodic payments the lessee makes to the lessor in addition to basic rent, "calculated with reference to the utilization of airframes, engines and other major life-limited components during the lease". They are set per flight hour or per flight cycle and collected per component: the airframe's heavy checks, the landing gear, the auxiliary power unit, each engine's performance restoration and each engine's life-limited parts. They pre-fund the eventual shop visit so that if the lessee defaults or hands the asset back, the lessor already holds cash for the life consumed. The lessor reimburses the lessee for a completed maintenance event up to the lesser of the reserve balance held for that event and the qualifying cost; any balance not claimed against a qualifying event stays with the lessor at lease end. AerCap calls them maintenance rents; Willis Lease books them as maintenance reserve revenue when they are no longer refundable. [AerCap 20-F FY2013; Spirit Airlines 10-K FY2013; WLFC Q3 2025 release]

END-OF-LEASE (EOL) COMPENSATION

End-of-lease compensation is the alternative regime, used for lessees whose credit is strong enough that the lessor does not demand monthly reserves. The lessee must "re-deliver the aircraft in a similar maintenance condition (normal wear and tear excepted) as when accepted under the lease, with reference to major life-limited components", and "to the extent that such components are redelivered in a different condition than at acceptance, there is an end-of-lease compensation adjustment for the difference at redelivery". The money can flow either way: the lessee pays if the asset comes back with less life than it had at delivery, and the lessor pays or credits if it comes back with more. Many leases combine the two regimes, reserves for some components and an end-of-lease adjustment for others. [AerCap 20-F FY2013]

The difference between the regimes is timing and credit, not amount. Under reserves the lessor collects monthly and holds the cash. Under end-of-lease compensation the lessor collects one lump at the end and bears the lessee's credit in the meantime. Either way the lessee pays for the life it consumes, measured against the condition at delivery, and §13 works both through the same lease.

12.4 RETURN CONDITIONS AND REDELIVERY

Part I §9 defined return conditions, also called redelivery conditions: the contractual minimum state of the asset at redelivery. For an engine that is typically a minimum number of flight hours

and cycles remaining to the next performance restoration, a minimum number of cycles remaining on the life-limited parts, a borescope inspection and a test-cell run. For an airframe it is time remaining to the next heavy check, fresh paint or a paint credit, and records in order. [definition assembled in the research, D6 §2.1] Redelivery is the act of handing the asset back at the end of the term, and the return conditions are what the lessor checks at that moment.

Return conditions do a specific job in the money flow. Together with the reserve or end-of-lease regime they decide who bears the cost of the maintenance life consumed during the lease. If the engine comes back below the required condition, the lessee must either restore it before redelivery or pay. Return conditions may also exclude parts and repairs that are not the manufacturer's; Part VI §42 carries that clause, and the research recorded that current lessor practice on it, for 2024 to 2026, was not found (D4 U1). [D4 §2.9; reconciliation C.4 item 2]

12.5 EXTENSIONS

A lease does not have to end at its term. The incumbent airline can renew it, and in the years this primer covers most did.

LEASE EXTENSION

A lease extension is the renewal of an expiring operating lease with the incumbent airline instead of taking the aircraft back. In 2023 to 2024, 75–90% of scheduled lease expiries on mid-life narrowbodies were being extended (Cirium Ascend, via Bloomberg); extension terms stretched to four to six years, and extension rents in some cases ran more than 40% above the age-specific rates agreed just before (IBA, November 2023). Part III §22 carries the full treatment. [Cirium Ascend via Bloomberg and AJOT (American Journal of Transportation), date not visible, context 2023–24; IBA Weekly, 24 Nov 2023; Key.Aero, "Lease trends"]

An extension matters here because it changes the return-condition arithmetic. The redelivery that would have triggered a settlement is postponed, the engines fly further toward their next shop visit, and the reserves or compensation keep accruing against a later date. IBA recorded cases in which aircraft due back at year twelve were not expected to return until 2032, at eighteen years of age. [IBA Weekly, 24 Nov 2023] Section 15 carries what that did to rents.

12.6 TRANSFERRING A LEASE: LEASE-ATTACHED TRADING AND NOVATION

A leased aircraft is often sold while the lease is running. The buyer wants the lease as much as the aircraft, because the lease is the cash flow.

LEASE-ATTACHED TRADING

Lease-attached trading is buying an aircraft from another lessor with the existing lease transferred to the buyer, so that the buyer acquires a rent stream and a return-condition obligation rather than an empty aircraft. The lease is what makes the asset financeable: lenders and securitization investors underwrite contracted rent from a known airline. Part VIII §60 and §67 cover FTAI's own transactions of this kind. [definition assembled in the research, D6 §2.4; D7 §2.4]

The legal step by which the lease moves is a novation: the old lessor, the new lessor and the airline agree that the new lessor takes the old lessor's place in the contract, with the airline's consent. No source in the research describes novation specifically; the word is used here in its ordinary legal sense to name the step the dossier describes as the lease being "transferred", and nothing in this primer turns on it. What does turn on it is that an on-lease aircraft changes hands with its deposit, its reserve balances and its return conditions attached, so the buyer inherits the whole money flow described in this section.

12.7 RESIDUAL VALUE

The last term is not in the contract but sits behind every number in it.

RESIDUAL VALUE AND RESIDUAL-VALUE RISK

Residual value is the lessor's forecast of what the asset will be worth, or fetch in a part-out, at the end of its depreciation life. Residual-value risk is the risk that the realised figure differs. Under an operating lease that risk stays with the lessor, which is what distinguishes it from a finance lease. FTAI's depreciation policy for leasing equipment, meaning the useful lives and residual assumptions it uses, was not returned by the research (D6 U8). [definition assembled in the research, D6 §2.1]

The parties who put numbers on residual value, and on current value, are the appraisers Part I §8 defined: IBA, mba Aviation, Ascend by Cirium and AVITAS (a US appraisal firm) publish base values and market values at half-life, and the market trades and the rating agencies test on their figures. Part I §8 also defined base value (the appraiser's opinion of the asset's underlying economic value in a balanced market) and market value (the most likely trading price in the actual current market, above base value when supply is tight and below it when soft). This Part uses those definitions without repeating them. What it adds is the observation that the research reached market values for the CFM56-7B (Exhibit 2.3) and no base values for any engine, and no appraised values of any kind for a 12-to-15-year-old 737-800 or A320ceo (D1 U9; D6 U9).

[IBA, Sept 2025; reconciliation cluster K-02]

EXHIBIT 2.1 — THE OPERATING LEASE, TERM BY TERM: WHAT EACH TERM DOES AND WHICH WAY THE MONEY FLOWS

Term	What it is	Who pays whom	Why it is in the lease	Source
Term	Fixed rental period; FTAI's remaining terms 44 months (aircraft), 38 months (engines) at 31 Dec 2025	n/a	Sets the period of contracted rent lenders underwrite	FTAI 10-K FY2025
Rent (lease rate)	Monthly payment; ~\$228,000 for a 12-yr 737-800 (Jul 2025)	Lessee to lessor	The lessor's primary income	IBA, Jul 2025
Lease rate factor	Rent ÷ market value, % per month	n/a	Gross-yield shorthand; no stated figure retrieved	Derived in the research
Security deposit	Cash or letter of credit at signing, returned on performance	Lessee to lessor, then back	Credit protection against default	D6 §2.1; FTAI 10-K (balance not returned)
Maintenance reserves	Per-hour and per-cycle payments by component; reimbursed up to the lesser of balance and qualifying cost	Lessee to lessor; lessor reimburses events	Pre-funds the life consumed	AerCap 20-F FY2013; Spirit 10-K FY2013
End-of-lease compensation	Cash adjustment at redelivery for the difference in maintenance condition	Either direction	Same purpose, settled at the end	AerCap 20-F FY2013
Return conditions	Minimum hours and cycles to next restoration, minimum LLP cycles, borescope, test run; airframe time to check, paint, records	Lessee restores or pays	Decides who bears the life consumed	D6 §2.1 (Part I §9)
Extension	Renewal with the incumbent; 75–90% of expiries extended, 2023–24	n/a	Postpones redelivery; rents >40% above prior rates in cases	Cirium Ascend; IBA, Nov 2023
Novation / lease-attached sale	Transfer of the lease to a new owner with the airline's consent	n/a	Makes an on-lease aircraft a tradeable cash flow	D6 §2.4 (novation not sourced in the research)

Term	What it is	Who pays whom	Why it is in the lease	Source
Residual value	Lessor's forecast end-of-life value	n/a	The risk the lessor keeps	D6 §2.1; FTAI policy not returned

Source: FTAI 10-K FY2025; IBA via India Seatrade News, 6 Sep 2026; AerCap 20-F FY2013; Spirit Airlines 10-K FY2013; Cirium Ascend via Bloomberg/AJOT; IBA Weekly, 24 Nov 2023; definitions assembled in D6 §2.1 and §2.4. Accessed 2026-10-03.

13. Maintenance reserves, return conditions and the lease-end settlement (worked example)

The maintenance terms are where most of the money in a mid-life lease actually sits. The reason is arithmetic. A performance restoration on a CFM56 costs between \$1.2M and \$2.5M on the two consultancy ranges Part I carried, and more with life-limited parts; an aircraft carries two such engines; and a six-year lease consumes a large fraction of a restoration interval on each. So the payments for life consumed can approach half of the basic rent, and the settlement at the end can be a multi-million-dollar negotiation. This section explains the mechanism, then works it through the one example the research built.

13.1 HOW A RESERVE IS SET AND SPENT

A reserve is collected by component because each component has its own clock. The airframe has heavy checks on a calendar and flight-hour schedule. The landing gear has an overhaul interval. The auxiliary power unit has its own hours. Each engine has two clocks of its own: the performance-restoration clock, measured in hours or cycles since the last restoration, and the life-limited-part clock, measured in cycles against each part's certified limit (Part I §4 and §5).

[mba Aviation, "Maintenance Matters in Aircraft ABS Deals", undated; ABS is the asset-backed securitization §17 defines; Maheshwari & Co., undated]

AUXILIARY POWER UNIT (APU)

The auxiliary power unit is the small gas turbine in the aircraft's tail that supplies electrical power and compressed air on the ground and in emergencies. It is one of the components a lease reserves for separately, and one of the higher-value items when an airframe is parted out. Part V §32 carries the part-out treatment. [D3 §2.2; D6 §2.1]

The rate for each component is set so that the balance accumulated by the next event roughly covers it. For an engine restoration the rate is per flight hour; for life-limited parts it is per flight cycle, because the parts are certified in cycles (Part I §4). No source the research reached gives

the rates a lessor actually charges on a CFM56 lease; the example below uses rates labelled as assumed, and the research recorded the gap (D6 U7; reconciliation C.2 item 6). [D6 §2.2]

Spending the reserve follows one rule, quoted from a lessee's own filing: the lessor reimburses "the lesser of (1) the amount of the maintenance reserve held by the lessor associated with the specific maintenance event or (2) the qualifying costs related to the specific maintenance event". [Spirit Airlines 10-K FY2013] Two consequences follow. If the event costs more than the balance, the lessee funds the difference. If the event costs less, the lessor keeps the rest. And because the reserve covers only life consumed since the lease began, not life consumed by earlier operators, a reserve rarely covers a whole shop visit unless the engine was delivered fresh from one. [D6 §2.2]

For the lessor, reserve receipts are not income when they arrive. They are a liability, carried as maintenance deposits, until the event is reimbursed or the lease ends with the balance unclaimed, at which point the unclaimed balance becomes revenue. [WLFC Q3 2025 release; D11 §2.2] The size of that line at an engine lessor shows how much of the money is maintenance rather than rent. Willis Lease reported maintenance reserve revenue of \$76.1M in the third quarter of 2025 alone, up 52.8% on a year earlier, against lease rent of \$76.6M in the same quarter; for the full year 2025 its combined "core lease rent and maintenance reserve revenues" were \$523.6M, of which lease rent was \$291.6M, leaving \$232.0M of maintenance reserve revenue, a figure derived in the research by subtraction. [WLFC Q3 2025 release; WLFC 10-K FY2025; derived in D11 §2.2] At AerCap, whose fleet is young, "maintenance rents and other receipts" were \$690M in 2025 against basic lease rents of \$6,679M, about 9%. [AerCap Q4 2025 release] The gap between 9% and 44% is the gap between a young fleet far from its shop visits and a mid-life engine book close to them. [D11 §2.2]

13.2 THE TWO REGIMES, SIDE BY SIDE

EXHIBIT 2.2 — MAINTENANCE RESERVES AGAINST END-OF-LEASE COMPENSATION: HOW EACH REGIME HANDLES THE SAME EVENTS

Event	Reserve regime	End-of-lease compensation regime	Source
Lessee flies 250 hours in a month	Pays rent plus a per-hour restoration reserve and a per-cycle LLP reserve for each engine	Pays rent only; the life consumed is tallied for settlement at redelivery	AerCap 20-F FY2013; D6 §2.2
Lessee performs a shop visit mid-lease	Claims reimbursement up to the lesser of the balance and the qualifying cost; funds any excess	Funds the visit; the engine's improved condition is credited at redelivery	Spirit 10-K FY2013; AerCap 20-F FY2013

Event	Reserve regime	End-of-lease compensation regime	Source
Lessee defaults mid-lease	Lessor holds the accrued balances and the security deposit	Lessor holds only the security deposit; the maintenance claim is unsecured	D6 §2.1
Redelivery below the return condition	Lessor keeps unclaimed balances and pursues any shortfall; lessee may restore before return	Lessee pays the agreed rate per hour and per cycle for life consumed below delivery condition	D6 §2.2; AerCap 20-F FY2013
Redelivery above the delivery condition	Balances are normally non-refundable; the lessee has pre-paid for life it did not consume	Lessor pays or credits the lessee for the excess life	D6 §2.1; AerCap 20-F FY2013
Who the regime suits	Lessees whose credit the lessor does not want to carry	Stronger-credit lessees; many leases mix the two by component	AerCap 20-F FY2013

Source: AerCap Holdings N.V., Form 20-F FY2013 (maintenance rents and end-of-lease compensation definitions); Spirit Airlines, Form 10-K FY2013 (lessee-side reserve accounting); mechanics assembled in D6 §2.1–2.2. Accessed 2026-10-03.

13.3 THE WORKED EXAMPLE

The research built one example of a full lease with reserves and a lease-end settlement. Its only sourced input is the rent. Every other input is an assumption chosen to show the arithmetic, and the shop-visit cost it assumed, \$4.5M, sits above every sourced range Part I carried, so the example below shows the sourced alternatives beside it. [D6 §2.2; reconciliation R-013, E.1 rule 4]

WORKED EXAMPLE 2.2 — A 12-YEAR-OLD 737-800 ON A SIX-YEAR OPERATING LEASE WITH MAINTENANCE RESERVES, AND THE SETTLEMENT AT THE END

Inputs. Rent \$228,000 a month (sourced: IBA, July 2025). Term 72 months (assumed). Utilization 250 flight hours (FH) and 140 flight cycles (FC) a month per engine (assumed; that is 1,680 cycles a year, a little above the 1,000–1,500 cycles a year Part I uses for a mature narrowbody from Aircraft Value News, and kept as the research set it). Performance-restoration reserve \$150 per flight hour per engine and life-limited-part reserve \$100 per flight cycle per engine (assumed; no sourced rates, D6 U7). One performance restoration on engine 1 in month 48 (assumed timing). Its cost: \$4.5M as the research assumed, shown beside the two consultancy ranges Part I carries for a restoration, \$1.8–2.5M (midpoint \$2.15M) and \$1.2–1.6M (midpoint \$1.4M), all from Safe Fly Aviation, a consultancy blog, 2026. At delivery both engines had flown 8,000 hours since their last restoration, and the return condition caps redelivery at 12,000 hours since restoration (both assumed).

Step 1, basic rent over the term.

$$228,000 \times 72 = \$16,416,000$$

Step 2, reserves accrued per engine over 72 months, then for two engines.

$$\begin{aligned} \text{restoration reserve: } & 150 \times 250 \times 72 = \$2,700,000 \text{ per engine} \\ \text{LLP reserve: } & 100 \times 140 \times 72 = \$1,008,000 \text{ per engine} \\ \text{two engines: } & 5,400,000 + 2,016,000 = \$7,416,000 \\ \text{reserves as a share of rent: } & 7,416,000 \div 16,416,000 = 45\% \end{aligned}$$

So on these assumptions the maintenance cash flow is not a side-show; it is almost half the size of rent.

Step 3, the month-48 shop visit on engine 1. The restoration balance held for that engine at month 48 is $150 \times 250 \times 48 = \$1,800,000$. The lessor reimburses the lesser of the balance and the qualifying cost.

$$\begin{aligned} & \text{at the research's assumed cost of } \$4,500,000: \text{ lessor reimburses} \\ & \quad 1,800,000; \text{ lessee funds } 2,700,000 \\ & \text{at the } \$2,150,000 \text{ midpoint of the } \$1.8\text{--} \$2.5\text{M range: lessor reimburses} \\ & \quad 1,800,000; \text{ lessee funds } 350,000 \\ & \text{at the } \$1,400,000 \text{ midpoint of the } \$1.2\text{--} \$1.6\text{M range: lessor reimburses} \\ & \quad 1,400,000; \text{ lessor keeps } 400,000 \end{aligned}$$

The three lines show the "lesser of" rule doing its work. At either of the higher costs the lessee pays the excess, because the reserve covers only the 48 months of life consumed under this lease, not the 8,000 hours consumed before delivery. At the lowest cost the reserve more than covers the visit and the lessor keeps the surplus. After month 48, engine 1 accrues restoration reserves again from zero.

Step 4, lease end on engine 2, which had no shop visit. It was delivered at 8,000 hours since restoration and flew $250 \times 72 = 18,000$ hours, so it would be at 26,000 hours since restoration. The return condition caps redelivery at 12,000. The lessee must therefore either perform a restoration before redelivery, claiming up to its \$2,700,000 balance against the cost, or negotiate a cash settlement. Under a pure reserve regime, if the lessee redelivers out of condition the lessor keeps the \$2,700,000 restoration balance and the \$1,008,000 LLP balance and pursues the shortfall. Under an end-of-lease compensation regime with no reserves paid, the lessee would owe compensation for 18,000 hours of restoration life consumed at an agreed rate per hour, plus compensation for $140 \times 72 = 10,080$ cycles of life-limited-part life at an agreed rate per cycle, in one lump at redelivery instead of monthly.

Step 5, what the lessor saw in cash over the lease, at the research's assumed visit cost.

$$\begin{aligned} \text{rent received: } & \$16,416,000 \\ \text{reserves received: } & \$7,416,000 \\ \text{reimbursed: } & -\$1,800,000 \\ \text{net reserve retention: } & 7,416,000 - 1,800,000 = \$5,616,000 \end{aligned}$$

That \$5.6M is not profit. It is the lessor's compensation for the maintenance life that walked out of the engines during the lease, to be spent, or forgone, when the lessor or the next lessee opens them. At the \$1.4M visit cost the retention would be \$6.0M, because less was reimbursed, and the lessee would have paid for more life than it used on engine 1.

Limits of the example: only the rent is sourced; the reserve rates, utilization, visit timing and return condition are assumed; the \$4.5M visit cost sits above every sourced range and the two consultancy ranges disagree with each other; the engine's life-limited-part reserve is never spent here because no life-limited part is replaced, which is the common case within a single six-year lease on parts with 20,000-to-30,000-cycle limits. Which set of life-limited-part prices a lessor uses to size that reserve, \$4M or \$5.7M for a full -7B set, is the tension Part I §4 carried (R-016). [D6 §2.2; IBA, Jul 2025; Safe Fly Aviation, 2026; Spirit Airlines 10-K FY2013; reconciliation R-013, R-016]

The example also shows why the maintenance condition at delivery is negotiated as hard as the rent. The lessee's exposure on engine 2 arose because the engine was delivered with 8,000 hours already consumed and the return condition was tighter than the delivery condition. A lessee who takes an engine fresh from a restoration and returns it at the same condition owes nothing; one who takes it part-worn and must return it fresh is buying the previous operator's consumption. In practice the delivery and return conditions are written to match, and the reserve or compensation covers the difference.

13.4 WHERE A LESSOR'S RETURN COMES FROM

A lease generates rent and reserves. The lessor's return is what is left after the asset's loss of value and the cost of the money that funded it. The research set out the decomposition in four lines.

THE FOUR LINES OF A LESSOR'S RETURN

Over the holding period a lessor earns (a) rental yield, meaning rent received, which is the lease rate factor times value, plus any net maintenance-reserve retention; minus (b) depreciation, the fall in the asset's value from purchase to disposal, or the book charge toward a residual; minus (c) financing cost, the interest on the debt that funds the asset and the cost of the equity; plus or minus (d) end-of-life proceeds, the sale price or part-out proceeds against the carrying residual, plus any end-of-lease compensation collected. An older asset has a higher lease rate factor and most of its depreciation behind it, but more maintenance exposure and a shorter remaining lease life, so line (d) matters more for it. In FTAI's structure line (d) is routed into the Aerospace Products segment: the 10 off-lease 737-700s bought from WestJet "will support FTAI's Aerospace Products business by expanding the Company's supply of CFM56-7B engines and modules" (Part VIII §67). [definition assembled in the research, D6 §2.1; FTAI release, 28 Sep 2026]

WORKED EXAMPLE 2.3 — DECOMPOSING A LESSOR'S RETURN INTO YIELD, DEPRECIATION, FINANCING COST AND RESIDUAL

Inputs, every one assumed for illustration and none of them market data: the aircraft is bought at value V ; the lease rate factor is 1.0% a month, so rent is 12% of V a year; straight-line depreciation is 3.5% of V a year toward a residual; debt funds 70% of V at 6.5%, so interest is $0.70 \times 6.5\% = 4.55\%$ of V a year; equity funds the other 30%. For a sense of scale only, the \$228,000 rent of worked example 2.2 at a 1.0% factor implies V of \$22.8M, which is not an appraised value (D6 U9).

rent yield: 12.00% of V
 less depreciation: -3.50% of V
 less interest: -4.55% of V
 net cash return on total capital: $12.00 - 3.50 - 4.55 = 3.95\%$ of V
 on the 30% equity slice: $3.95 \div 0.30 = 13.2\%$ of equity

Now change one input. At a lease rate factor of 0.8% a month, rent is 9.6% of V , and the net return falls to $9.6 - 3.5 - 4.55 = 1.55\%$ of V , or 5.2% on equity. At 1.2% a month it rises to $14.4 - 8.05 = 6.35\%$ of V , or 21.2% on equity. The leverage multiplies both directions: a 0.2-point move in the monthly factor moves the equity return by about eight points on these inputs.

Two further flows sit outside those lines. Net maintenance-reserve retention adds cash, \$5.6M over six years in worked example 2.2, which on V of \$22.8M is about 4.1% of V a year, but it is matched by the restoration cost the lessor will face before re-leasing; it is timing, not margin, unless the asset is parted out instead. End-of-life proceeds against the carrying residual add or subtract, and for an aircraft whose value sits mostly in its engines (§15) that line is largely the engines' value. Downtime between leases, remarketing costs and end-of-lease shortfalls subtract. Limits of the example: every percentage is assumed; FTAI's own depreciation policy and cost of debt by asset were not retrieved (D6 U8); Part VIII §58 works FTAI's implied gross yield from its filed balances and Part XI §88 sets it beside Willis Lease and Air Lease. [D6 §2.2 step 6; D6 §2.7]

The structure of that return explains the behaviour of the specialists this primer describes. A new-aircraft lessor earns a low factor on a slowly depreciating asset with long leases and little maintenance exposure. A mid-life engine lessor earns a higher factor on an asset whose value is close to the sum of its modules, with reserves nearly as large as rent and a part-out decision at every lease end. Section 14 puts filed numbers on both, and Part XI §88 sets FTAI's beside them.

14. Engine leasing: spare ratios, short-term leases, lease-rate factors

Everything in §12 and §13 applies to an engine leased on its own, with two differences. An engine is a smaller, more mobile asset than an aircraft, so its leases can be as short as a few months. And an engine's value is dominated by its maintenance condition, so the lease-end decision is sharper: restore and re-lease, lease out the remaining life, or take it apart. This section

explains why airlines lease engines at all, how the leases are priced, and what the one listed pure-play engine lessor's numbers look like.

14.1 WHY AIRLINES LEASE SPARE ENGINES

An airline needs more engines than it has engine positions, because engines come off the wing for scheduled shop visits and for unscheduled removals. Part I §6.4 defined the spare engine and the spare ratio, spares as a percentage of installed engines, and quoted the statement of the German engine maker MTU Aero Engines (MTU) that airlines "used to have about 15% of spare engines of their installed engines, but this figure is now down to 10% and is expected to further decrease to about 7–8% for newer engine types". [MTU AEROREPORT, undated] A consultancy argues the 10% rule "is no longer sufficient because engines are coming off wing more often and taking longer to repair". [Acumen Aviation, undated (consultancy blog)]

The economic reason to lease rather than own is that a spare earns nothing while it waits. Airlines "are less inclined to keep as many owned spare engines as they used to, mainly because of the significant investment involved in assets that generate very little return". [MTU AEROREPORT, undated] Leasing converts a lumpy multi-million-dollar purchase into a monthly cost, and in a lessor's pool one spare can serve several operators in turn. The sharpest form of the need has its own name.

AIRCRAFT ON GROUND (AOG)

An aircraft on ground is one that cannot fly for want of a serviceable part or engine. Short-term engine leases exist to cover AOG events and planned shop visits; a lessor that can deliver an engine within days is selling the airline the difference between a grounded aircraft and a flying one. [MTU AEROREPORT, undated; FlightGlobal, "Freedom to be flexible", undated]

14.2 SHORT-TERM AND LONG-TERM LEASES, AND SHOP-VISIT COVER

Part I §6.4 defined the two durations. A short-term engine lease runs "a period of a few months to a maximum of three years" and is "ideal if a replacement engine is required for the duration of a shop visit"; a long-term lease runs for years and provides a permanent spare. [MTU AEROREPORT, undated; FlightGlobal, undated] The short-term lease is the one that ties engine leasing to the shop-visit market Part IV describes. A CFM56 restoration takes an engine away for 75 to 120 days on the figures Part I §6.3 carried, and when shops are full the wait for a place in the queue adds two to three months, up to six. [Safe Fly Aviation, 2026 (consultancy blog); Aviation Business News, 2025; GE Aerospace, 27 May 2026; Bain, 2024; reconciliation R-019]

INDUCTION SLOT

An induction slot is the scheduled date on which an engine shop commits to start work on an engine; induction is the acceptance of the engine into work. When shops are full the wait for a slot is additional to the turnaround time and, in 2024, became the binding constraint, adding two to three months and up to six. Part IV §27 carries the full treatment. [Bain, 2024; D2 §2.1, §2.4]

When slots are delayed, "operators often turn to short-term or green-time engine leasing". [MTU AEROREPORT, undated] A green-time engine, which Part I §8.2 defined and §16 of this Part carries, is one with limited life left before its next shop visit, leased out to burn that life rather than be restored; its rent is lower. The lessor's product is therefore time: the months between removal and return, priced per month. FTAI's offer sits beside it as a different answer to the same problem.

MAINTENANCE, REPAIR AND EXCHANGE (MRE)

Maintenance, Repair and Exchange is FTAI's name for its offer, a deliberate contrast with the industry's maintenance, repair and overhaul (MRO). Instead of restoring a customer's engine part by part while the customer waits, or leasing the customer a spare for the duration, FTAI removes the module or engine that triggered the visit, installs a serviceable one from its inventory, tests and returns the engine, and takes the unserviceable unit as feedstock. It is a variant of shop-visit cover in which the exchange replaces the spare. Part VII §46 carries the full treatment. [FTAI 10-K FY2025; D5 §2.2; D6 §2.3]

WORKED EXAMPLE 2.4 — HOW MANY ENGINES A FLEET HAS IN THE SHOP, AND WHAT THE SPARE RATIO MUST COVER

Inputs: a fleet of 100 Boeing 737-800s, each with two CFM56-7B engines, so 200 installed engines (assumed fleet; two engines per aircraft sourced in Part I). Spare ratios of 15%, 10% and 7–8% (sourced: MTU). A mature performance-restoration interval of 10,000–15,000 cycles (sourced: Leeham News, 8 Mar 2024) at 1,000–1,500 cycles a year (sourced: Aircraft Value News), which is between 6.7 and 15 years between restorations; an average of ten years (assumed, close to the roughly ten years implied by 2,300 visits a year on about 24,000 engines, derived in the research) with eight years (Aviation Week's rule of thumb) and six years (the assumption one dossier used and asked to have replaced) as alternatives. Turnaround of 90 days (sourced: GE Aerospace, May 2026) or 120 days (sourced: trade press, 2025), with a slot wait of 60–90 days (sourced: Bain, 2024).

spares at 15%: $0.15 \times 200 = 30$; at 10%: 20; at 7–8%: 14–16
 scheduled removals a year at a 10-year interval: $200 \div 10 = 20$ (8 years: 25; 6 years: 33)
 engines away at any time = removals $\times$ days away $\div$ 365
 10-year interval, 90 days: $20 \times 90 \div 365 = 4.9$
 10-year interval, 120 days: $20 \times 120 \div 365 = 6.6$
 10-year interval, 120 days plus a 90-day slot wait: $20 \times 210 \div 365 = 11.5$
 8-year interval, 90 to 210 days: 6.2 to 14.4
 6-year interval, 90 to 210 days: 8.1 to 19.0

So on the ten-year interval, scheduled restorations alone keep between five and twelve of the fleet's 200 engines away at any moment, and the rest of a 20-engine spare pool (10%) is what covers unscheduled removals and grounded aircraft. Lengthen the time away from 90 to 210 days and the scheduled requirement alone more than doubles. Shorten the interval to the six years one dossier assumed and the requirement at 210 days nearly consumes the whole 10% pool before any unscheduled event. That is the arithmetic behind the consultancy's claim that 10% "is no longer sufficient", and behind the demand for short-term leases when slots are delayed. Limits of the example: the interval, the fleet and the averaging are assumed; unscheduled removal rates were not sourced; an engine exchange from inventory, which aims at a quick-turn of about 45 days, would cut the days-away figures by half or more and is the mechanism Part VII §55 works through. [MTU AEROREPORT, undated; Leeham News, 8 Mar 2024; Aircraft Value News, undated; GE Aerospace, 27 May 2026; Aviation Business News, 2025; Bain, 2024; Aviation Week, "Retirement Uptick", date not captured; reconciliation R-008, R-019; derived in D9 T1]

14.3 LEASE RATES, VALUES AND THE LEASE-RATE-FACTOR ARITHMETIC

The research reached two lease rates for a CFM56-7B that do not agree, and two kinds of value that sit on different bases. Part I §8.2 set them out in Exhibit 1.11 as values; this Part sets them out as rents and factors. The rule the primer follows is that both figures appear together with their sources, that no lease rate factor is stated as a single number, and that every derived factor carries its basis. [reconciliation R-028, R-029, R-030; E.1 rule 6]

EXHIBIT 2.3 — CFM56 AND V2500 (THE A320CEO'S ALTERNATIVE ENGINE, PART I §10) LEASE RATES AND VALUES, WITH THE LEASE RATE FACTOR EACH PAIR IMPLIES AND THE BASIS ON WHICH IT SITS

Quantity	Figure	Basis	Source and level	Date
CFM56-7B spare-engine rent	~\$100,000 a month, from ~\$75,000 in 2019 (+33%)	Spare engine; condition not stated	IBA (appraiser), via AviTrader and AJOT	Apr 2024

Quantity	Figure	Basis	Source and level	Date
CFM56-7B rent	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Quoted beside green-time values; may be green-time; basis not stated	Safe Fly Aviation (consultancy blog)	2026
CFM56-5B rent	\$38,000–44,000 a month	Same	Safe Fly Aviation	2026
V2500-A5 rent	\$70,000–80,000 a month; "slightly higher" than the -5B in 2024	Typical	IBA, Engine Values 2025B; IBA Apr 2024	Sept 2025; Apr 2024
CFM56-7B24 / -7B27 market value	\$5.7M / \$6.4M	Half-life, H2 2025	IBA, 2025B (appraiser)	Sept 2025
CFM56-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", internally inconsistent	Safe Fly Aviation	2026
Run-out CFM56 as teardown feedstock	\$0.8–1.2M	Exhausted engine	Safe Fly Aviation	2026
Value direction	"Levelled off after approximately 18 months of increases"	-5B and -7B market values	IBA H1 2026 update	H1 2026
Implied factor: IBA rent on IBA value	1.56% a month (-7B27); 1.75% (-7B24)	Rent 2024, value H2 2025; dates mixed	Derived in this Part	n/a
Implied factor: Safe Fly rent on IBA value	0.66–0.75% (-7B27); 0.74–0.84% (-7B24); 0.70% at the \$45,000 midpoint	Possibly green-time rent on half-life value; bases mixed	Derived in D3 §2.3 and this Part	n/a
Implied factor: Safe Fly rent on Safe Fly value	1.24–1.71% a month	Both possibly green-time; label inconsistent	Derived in this Part	n/a
Willis Lease portfolio, derived	~1.0% a month on the on-lease portion	\$291.6M rent ÷ \$2,801.7M equipment ÷ 84.9% utilisation	Derived in D11 §2.2 from WLFC 10-K FY2025	FY2025

Quantity	Figure	Basis	Source and level	Date
FTAI leasing book, derived	0.87% a month on cost, 1.1% on net book value (1H 2025 annualised); 0.55% and 0.73% on 1H 2026 income	Lease income ÷ leasing equipment; mixed aircraft and engines	Derived in D6 §2.7; Part VIII §58	FY2025 / 1H 2026
Assumption used elsewhere	0.9% a month	Assumed for a fee example	D7 §5.1; Part VIII §66	n/a

Source: IBA, "It's a lessors' market", Apr 2024, via AviTrader, 25 Apr 2024, and AJOT; IBA, Engine Values Release 2025B, Sept 2025; IBA, Engine and Lease Rate Update H1 2026; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); WLFC 10-K FY2025; FTAI 10-K FY2025 and 10-Q Q1 2026. Derivations in D3 §2.3, D6 §2.7, D11 §2.2 and this Part. Reconciliation R-028, R-029, R-030. Accessed 2026-10-03.

The two rents are a direct contradiction at face value and possibly a difference of definition. An appraiser quoted about \$100,000 a month in April 2024 for a spare engine; a consultancy blog quoted \$42,000 to \$48,000 in 2026 on a page that also quotes "green-time values", and the research noted that the lower rent may describe green-time engines although the source does not say so. [IBA, Apr 2024; Safe Fly Aviation, 2026; D6 T4] The two sources agree on one thing, the change since 2019: IBA's \$75,000 to \$100,000 is a rise of a third, and the blog's rate is "about 34% above 2019". [reconciliation B-5] They disagree on the level, and the primer carries both; they also disagree on direction, the appraiser's "levelled off" against the blog's rents up 15–20% "in recent months", which §16.4 sets out. [IBA, H1 2026; Safe Fly Aviation, 2026; reconciliation R-029]

WORKED EXAMPLE 2.5 — THE LEASE-RATE-FACTOR ARITHMETIC FOR A CFM56-7B SPARE ENGINE, ON BOTH RENTS

Inputs: rent of about \$100,000 a month (sourced: IBA, April 2024) or \$42,000–48,000 a month (sourced: Safe Fly Aviation, consultancy blog, 2026; may be green-time). Half-life market value \$6.4M for a -7B27 and \$5.7M for a -7B24 (sourced: IBA, September 2025). "Green-time value" \$2.8–3.4M (sourced: Safe Fly Aviation, 2026; label internally inconsistent). The research's own derivation uses the \$45,000 midpoint (D3 §2.3).

$$\begin{aligned}
 &\text{IBA rent on IBA value:} \\
 &100,000 \div 6,400,000 = 1.56\% \text{ a month} = 18.8\% \text{ a year} \quad (-7B27) \\
 &100,000 \div 5,700,000 = 1.75\% \text{ a month} = 21.1\% \text{ a year} \quad (-7B24) \\
 &\text{Safe Fly rent on IBA value:} \\
 &42,000 \div 6,400,000 = 0.66\%; \quad 48,000 \div 6,400,000 = 0.75\% \quad (-7B27) \\
 &42,000 \div 5,700,000 = 0.74\%; \quad 48,000 \div 5,700,000 = 0.84\% \quad (-7B24) \\
 &45,000 \div 6,400,000 = 0.70\% \text{ a month} = 8.4\% \text{ a year} \quad (\text{the research's figure}) \\
 &\text{Safe Fly rent on Safe Fly "green-time value":} \\
 &42,000 \div 3,400,000 = 1.24\%; \quad 48,000 \div 2,800,000 = 1.71\% \text{ a month}
 \end{aligned}$$

Read line by line. The first pair mixes a 2024 rent with a late-2025 value, and a spare engine's maintenance condition is not stated, so 1.56–1.75% is indicative, not measured; it is the highest factor in this primer. The second group mixes what may be a green-time rent with a half-life value, which the research itself flagged as a mixed basis, and it gives the lowest factors, 0.66–0.84%. The third pair puts the blog's rent on the blog's own value and gives 1.24–1.71%, close to the first pair, which is what one would expect if both the rent and the value describe a part-worn engine: a cheaper engine renting for less at a similar factor. None of this resolves the contradiction between the two rents; it shows that the factor depends entirely on which rent is set against which value. For comparison, Willis Lease's derived portfolio factor is about 1.0% a month on the on-lease portion of its book, and FTAI's derived book factor is 0.87–1.1% a month on 2025 income, both on mixed books of engines and aircraft at varying condition. Limits of the example: every input is contested or mixed in date or basis; no stated lease rate factor exists in the research for any party (D6 U5; D11 U1); and the V2500-A5 rent of \$70,000–80,000 cannot be turned into a factor because no V2500 value was retrieved. [IBA, Apr 2024; IBA, Sept 2025; Safe Fly Aviation, 2026; derived in D3 §2.3, D6 §2.7, D11 §2.2; reconciliation R-028, R-029, R-030]

GROSS LEASING YIELD

Gross leasing yield is lease income for a period divided by the average leasing equipment at cost over the period, or at net book value (cost less accumulated depreciation). It is the filed-accounts counterpart of the lease rate factor: a company does not state its factor, but its income statement and balance sheet give the yield, and dividing by twelve gives a factor on book rather than on market value. Willis Lease's FY2025 figure is about 10.4% on year-end equipment, or 12.3% on the on-lease portion; Air Lease's, a new-aircraft lessor's, is about 9.0% (\$2,615.4M of rental revenue on \$29.1B of flight equipment at 31 December 2025); FTAI's FY2025 figure is about 10.4% on average cost and 13.4% on average net book value using first-half 2025 income annualised. All three are derived in the research on year-end or average book rather than market value, so they approximate the companies' own measures. Part VIII §58 and Part XI §88 reuse the measure. [derived in D11 §2.2 and D6 §2.7; WLFC 10-K FY2025; Air Lease 10-K FY2025; FTAI 10-K FY2025]

14.4 WILLIS LEASE AS THE REFERENCE

Willis Lease Finance Corporation is the only US-listed company whose primary business is engine leasing, which makes its filings the closest public analogue to an engine book like FTAI's. [D11 §3.1] It owns engines and a small number of aircraft, leases them to airlines and maintenance shops, collects maintenance reserves, sells engines and parts out of the portfolio, and manages engines for third parties. Its numbers appear in this primer at two dates, a quarter-end and a year-end, and the two are not contradictions but different measures; both are quoted because both will be used. [reconciliation R-032]

EXHIBIT 2.4 — WILLIS LEASE FINANCE: THE ENGINE-LESSOR METRICS AT TWO DATES

Metric	At or for 30 Sep 2025 (quarter)	At or for 31 Dec 2025 (year)	Note	Source
Utilisation	86.0% (82.9% a year earlier)	84.9% average for 2025 (82.9% for 2024)	Point-in-time against period average; by book value on lease	WLFC Q3 2025 release; WLFC 10-K FY2025
Lease portfolio	\$2,888.5M, of which \$2,700.4M equipment in operating leases, \$144.8M notes receivable, \$27.0M maintenance rights, \$16.3M sales-type leases	\$2,801.7M equipment held for lease; \$139.9M notes receivable; \$30.6M maintenance rights; \$16.6M sales-type leases	"Maintenance rights" is a balance-sheet line the research did not define (Part XI §87)	Same
Including joint ventures	\$3,302.6M	Not returned		WLFC Q3 2025 release
Owned assets	Not returned	363 engines, 20 aircraft, one marine vessel; 69 lessees in 37 countries		WLFC 10-K FY2025
Third-party managed	Not returned	116 engines and related equipment	The analogue of FTAI's servicing of its vehicles (Part VIII)	WLFC 10-K FY2025
Lease rent revenue	\$76.6M in Q3 (+17.9%, a record)	\$291.6M (+22.4% from \$238.2M)		Same
Maintenance reserve revenue	\$76.1M in Q3 (+52.8%, a record)	\$232.0M, derived as \$523.6M combined line less \$291.6M rent	Reserves recognised when no longer refundable	Same; derived in D11 §2.2
Core lease rent plus maintenance reserve revenue	Not returned	\$523.6M (+15.8% from \$452.1M)	Reserves are 44% of the combined line	WLFC 10-K FY2025
Gain on sale of leased equipment	Not returned	\$54.0M (+19.9%)	Where the sum-of-modules value is realised	WLFC 10-K FY2025

Metric	At or for 30 Sep 2025 (quarter)	At or for 31 Dec 2025 (year)	Note	Source
Total revenue	Not returned	\$730.2M (+28.3%)		WLFC 10-K FY2025
Lease rate factor	Not disclosed	Not disclosed; ~1.0% a month derived	D6 U5; D11 U1	Derived in D11 §2.2

Source: Willis Lease Finance Corp., Q3 2025 earnings release (8-K exhibit 99.1, Nov 2025) and Form 10-Q for the quarter ended 30 Sep 2025; Willis Lease Finance Corp., Form 10-K FY2025. Derivations in D11 §2.2. Reconciliation R-032. Accessed 2026-10-03.

Three things in that table carry into the rest of the primer. First, utilisation is a defined term and the definitions differ by company: Willis measures the share of the lease portfolio, by book value, that is on lease over the period; FTAI measures the percentage of days on lease in the quarter, weighted by the monthly average equity value of its leasing equipment and excluding airframes. The two are not comparable without saying so, and Part VIII §58 and Part XI §88 say so where they set 84.9% beside FTAI's roughly 77%. [WLFC 10-K FY2025; FTAI 10-K FY2025; registry C-3] Second, maintenance reserve revenue at a mid-life engine lessor is of the same order as rent, 44% of the combined line for the year and above rent in one quarter, because the assets are near their shop visits; this is the §13 mechanism showing up in a filing. Third, the gain on sale, \$54.0M in 2025, is the line on which a lessor realises the sum-of-modules value Part I §8 described, and Part V §32 carries what a part-out yields. [WLFC 10-K FY2025; D11 §2.2]

Willis funds itself through a series of rated engine securitizations, the Willis Engine Structured Trust (WEST) deals, which §17 uses as the sourced template for how an engine portfolio is financed. Its lease rate factor, its split between CFM56 and newer engine types, and its spare-parts revenue line were searched for and not returned (D11 U1, U3; D6 U5). [D11 §3.1]

14.5 THE OTHER ENGINE LESSORS

The research named the other participants and retrieved scale for few of them. The list is carried as the research left it, with what was and was not found.

EXHIBIT 2.5 — ENGINE LESSORS NAMED IN THE RESEARCH, AND WHAT WAS RETRIEVED FOR EACH

Lessor	Ownership or affiliation	What the research retrieved	Source
Willis Lease Finance	Listed pure-play	363 engines, 20 aircraft, 116 managed, 84.9% utilisation, \$291.6M lease rent (FY2025); Exhibit 2.4	WLFC 10-K FY2025

Lessor	Ownership or affiliation	What the research retrieved	Source
GE Engine Leasing	GE Aerospace, co-owner of CFM International, the CFM56's original equipment manufacturer (OEM)	A 2019 trade report: "has seen an explosion of financing based on spare engines"; engine leasing "a key part of full-service package deals" from the engine makers; no current scale	AviTrader MRO e-magazine, Jul 2019
AerCap Engines; Shannon Engine Support (SES)	AerCap; SES is AerCap's 50% joint venture with Safran Aircraft Engines	470 owned engines, 9% of AerCap's fleet, at 31 Dec 2025 (20-F); "over 1,200 engines" including engines owned and managed by SES at 30 Sep 2025 (6-K); \$155M of lease rental income from SES in nine months of 2025	AerCap 20-F FY2025; AerCap 6-K Q3 2025
SMBC Aero Engine Lease (SMBC)	Sumitomo Mitsui group, affiliated with SMBC Aviation Capital	Scale not retrieved	D6 §3
Engine Lease Finance (ELF)	Mitsubishi HC Capital; Shannon-based	Scale not retrieved	D6 §3
Rolls-Royce & Partners Finance (RRPF)	Joint venture of Rolls-Royce and GATX Corporation (GATX)	2019 report: market expansion from "airlines becoming more open to spare engine leasing, driven by lower lease rentals in an increasingly competitive market"; no current scale	AviTrader, Jul 2019
MTU Maintenance Lease Services	The leasing arm of MTU's maintenance business	"Short-term leasing, stand-by arrangements, engine pooling, as well as asset management"; no scale	MTU AEROREPORT, undated
FTAI Aviation	Listed; Aviation Leasing segment	243 engines at 31 Dec 2025, 143 on lease; Part VIII §58	FTAI 10-K FY2025
Top five engine lessors together	n/a	"Over 1,000 powerplants with a book value in excess of \$5 billion"	AviTrader, Jul 2019 (dated)

Source: WLFC 10-K FY2025; AviTrader MRO e-magazine, July 2019 (via AJW Group); AerCap Holdings N.V., Form 20-F FY2025 and Form 6-K for the quarter ended 30 Sep 2025; MTU AEROREPORT, "How airlines benefit from engine leasing", undated; FTAI 10-K FY2025. Scale for the unlisted lessors was not retrieved (D6 U14). Accessed 2026-10-03.

Two cautions attach to the table. AerCap's engine count is 470 or more than 1,200 depending on whether the figure is owned engines only or owned plus managed plus the Shannon joint

venture's, and the two dates are a quarter apart; the primer writes "470 owned" unless it states the wider basis. [AerCap 20-F FY2025; AerCap 6-K Q3 2025; reconciliation R-033] And the only figure for the group as a whole, more than 1,000 engines and more than \$5 billion of book value among the top five, is from 2019 and is carried as dated. [AviTrader, Jul 2019; D6 U14] The point that survives the gaps is structural: the engine makers themselves, the largest aircraft lessor, two bank-affiliated specialists, an independent repair shop and one listed pure-play all lease spare engines, and the pure-play's filings are the only place the economics are laid out in full.

15. Mid-life narrowbody economics: why the engines carry the value

A mid-life narrowbody is the aircraft at the centre of this primer: the Boeing 737NG and the Airbus A320ceo, powered by the CFM56-7B and CFM56-5B that Part I described, in the second half of their service lives. Part I §4 boxed the term with both of the research's framings, and they are consistent: roughly 10 to 18 years old (one dossier), or roughly the middle of a 25-to-30-year economic life (another), when an increasing share of the aircraft's value sits in its engines and those engines are approaching or past a shop visit. [D6 §2.4; D7 §2.4; registry C-13] This section explains what has happened to those aircraft's values and rents since 2023, why they are bought with leases attached, why the engines carry the value, and what happens to the airframe at the end.

15.1 VALUES AND LEASE RATES, 2023 TO 2026

The research reached commentary and lease rates from one appraiser at several dates, a lessor's filing, and a data provider's remarks via the financial press. It did not reach an appraised half-life value for a 12-to-15-year-old 737-800 or A320ceo at any date (D6 U9). The sourced points are set out with their dates because the direction changed within the period.

EXHIBIT 2.6 — MID-LIFE NARROWBODY VALUES AND LEASE RATES, 2023 TO 2026: EVERY SOURCED POINT WITH ITS DATE AND LEVEL

Date	Statement or figure	Asset	Source and level
Context 2023–24	75–90% of scheduled lease expiries being extended; only about a dozen mid-life 737NGs and fewer than 25 A320ceos available to rent worldwide	Mid-life narrowbodies	Cirium Ascend (Rob Morris) via Bloomberg/AJOT (data provider via financial press; date not visible)
Context 2023–24	Lease rates on 10-year-old 737-800s 44% above January 2022 levels	10-yr 737-800	Cirium Ascend via Bloomberg/AJOT

Date	Statement or figure	Asset	Source and level
Nov 2023	Extension terms stretched to 4–6 years; extension rents in some cases >40% above the prior age-specific rate; aircraft due back at year 12 not expected until 2032, at age 18	Legacy single-aisles	IBA Weekly, 24 Nov 2023; Key.Aero (appraiser)
Feb 2024	"Values and leases would continue to soar throughout the year, attributed to tight supply driven by lease extensions, engine reliability issues, and a lack of new aircraft deliveries"	Narrowbodies	IBA, Feb 2024 update (appraiser), as summarised in search
Apr 2024	CFM56-7B saw the largest market-value gain among engines covered, "around 20% from 2023 to 2024"; "a lessor's market"	-7B engine	IBA via AviTrader, 25 Apr 2024 (appraiser)
Jul 2025	12-year-old 737-800 about \$228,000 a month (–11% year on year); comparable A320-200 about \$220,000 (–13%); 21-year-old 737-800s "less than \$160,000"; older end "comfortably sub-\$200k"	Mid-life and old narrowbodies	IBA, Jul 2025, via India Seatrade News and AeroIn (appraiser)
Jul 2025	"Greater availability in the secondary market and softening operator demand" pressing lease rates down; "strong demand for engines and component scarcity should support base aircraft values despite the weaker leasing environment"	Narrowbodies	IBA, Jul 2025 (appraiser)
FY2025 (filed early 2026)	Strong demand "helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong as the supply and demand environment for commercial aircraft remains tight"	Air Lease's fleet (new-technology, average age 4.9 years)	Air Lease 10-K FY2025 (filing)
FY2025 (filed early 2026)	"Both Airbus and Boeing have delivery delays which have been compounded by engine manufacturer delays, shorter on-wing engine time of most new technology engines, and ongoing supply chain challenges"	Engines	WLFC 10-K FY2025 (filing)

Date	Statement or figure	Asset	Source and level
H1 2026	-5B and -7B values "have levelled off after approximately 18 months of increases"; "early signs of stabilisation in parts of the narrowbody segment"	Engines	IBA H1 2026 update (appraiser)
6 Sep 2026	IBA "warns of oversupply of narrowbody conversions and lease rate fall"	Narrowbodies; freighter conversions	IBA via India Seatrade News (headline)
2025 (date not captured)	Freighter values rising while narrowbody passenger lease rates receded	Freighters against passenger narrowbodies	IBA via Aeroin (appraiser via trade press)

Source: Bloomberg via AJOT, "Used jets rents surge 44% as airlines pay up to buoy fleets"; IBA Weekly, 24 Nov 2023; Key.Aero, "Lease trends"; IBA, "Aircraft values and lease rates update, February 2024"; AviTrader, "IBA says it's a lessor's market", 25 Apr 2024; India Seatrade News, "IBA warns of oversupply of narrowbody conversions and lease rate fall", 6 Sep 2026; Aeroin, 2025; Air Lease Corp., Form 10-K FY2025; Willis Lease Finance Corp., Form 10-K FY2025; IBA, Engine and Lease Rate Update H1 2026. Reconciliation R-031. Accessed 2026-10-03.

The table holds a divergence the primer carries rather than resolves. A new-aircraft lessor's filing for 2025 says lease rates rose during the year and are expected to remain strong; an appraiser's mid-life commentary from July 2025 has 12-year-old 737-800 rents down 11% year on year and expects more softening; and the same appraiser warned in September 2026 of conversion oversupply and falling rates. [Air Lease 10-K FY2025; IBA, Jul 2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-031] The sources describe different asset ages, Air Lease's fleet averaging 4.9 years against the appraiser's 12-to-21-year-old aircraft, and different dates, so the divergence may be a difference of segment and timing rather than a contradiction. [D11 §3.3; D9 §2.8] The appraiser's own July 2025 sentence contains the mechanism this section is about: rents on the whole aircraft were softening while "strong demand for engines and component scarcity" supported the aircraft's base value. The two forces pull a mid-life lessor's rent and its engine-exit value in opposite directions. [IBA, Jul 2025; D6 §5]

Behind the rent movements sits the extension behaviour §12.5 defined. If 100 leases on A320neos expire in a year and 75 to 90 are extended, only 10 to 25 aircraft come back to lessors for transition, sale or part-out, and that is the whole flow into the used market from that cohort; with fewer than 25 A320neos available to rent worldwide at the time of Cirium's count, the stock of unplaced aircraft was of the same order as one year's returns from a 100-aircraft cohort, which is the arithmetic behind a 40%-plus rent move. [derived in D9 §2.8; Cirium Ascend via Bloomberg/AJOT] Part III §22 carries the extension story in full and Part III §24 turns it into a row of the translation table.

15.2 WHY MID-LIFE AIRCRAFT ARE BOUGHT WITH A LEASE ATTACHED

A buyer of a 12-year-old aircraft almost always buys it with a lease already on it. There are two routes. In the first, the operating airline sells aircraft it owns to a lessor and rents them straight back, a sale-leaseback, which §17.7 defines; FTAI's September 2026 WestJet transaction, in which a vehicle bought 17 Boeing 737-700s "on lease to WestJet in a sale-leaseback", is the example in the research and belongs to Part VIII §67. [FTAI release, 28 Sep 2026] In the second, lease-attached trading (§12.6), the buyer takes the aircraft from another lessor with its lease transferred. [D6 §2.4]

The lease is what makes the asset financeable. Lenders and securitization investors underwrite contracted rent from a known airline for a known term, and the buyer's return over that term is the decomposition of worked example 2.3. [D6 §2.4] At the end of the term the owner has three choices: re-lease the aircraft, convert it to a freighter, or part it out. The value of that choice is mostly the engines, and the next subsection shows why with the one sourced illustration the research built.

15.3 WHY THE ENGINES CARRY THE VALUE

WORKED EXAMPLE 2.6 — TWO CFM56-7B RENTS AGAINST A 737-800 RENT, ON BOTH ENGINE RENTS

Inputs: a CFM56-7B spare engine rented for about \$100,000 a month in 2024 (sourced: IBA, April 2024), or for \$42,000–48,000 a month in 2026 (sourced: Safe Fly Aviation, consultancy blog; may be green-time); a complete 12-year-old 737-800 rented for about \$228,000 a month in July 2025 (sourced: IBA). A 737-800 carries two engines (sourced in Part I).

on the IBA engine rent:
 two engines: $2 \times 100,000 = \$200,000$ a month
 share of the aircraft's rent: $200,000 \div 228,000 = 88\%$
 left for airframe, landing gear, APU and everything else: $228,000 - 200,000 = \$28,000$ (12%)
 on the Safe Fly engine rent:
 two engines: $2 \times 42,000$ to $2 \times 48,000 = \$84,000$ to $\$96,000$ a month
 share of the aircraft's rent: 37% to 42%
 left for everything else: $\$132,000$ to $\$144,000$ (58% to 63%)

On the appraiser's engine rent, two engines on standalone leases would command 88% of what the whole aircraft commands, and everything else about 12%. On the consultancy blog's rent the engines are 37–42% and the rest 58–63%. The primer carries both. The first calculation mixes an April 2024 engine rent with a July 2025 aircraft rent, and the appraiser's own H1 2026 update says engine values "levelled off", so the 88% is indicative rather than measured; the research flagged exactly this (D6 T5). The second calculation uses a rent that may price a part-worn engine, which would understate what a full-life spare earns. What neither calculation disputes is the direction: a pair of engines earns

a large share of a mid-life narrowbody's rent on either figure, and a pair of fresh ones earns most of it on the appraiser's.

The same comparison in values cannot be completed from the research. Part I's worked example 1.7 put a full-life -7B27 at about \$10.3M on one set of inputs or \$9.1M on the alternative, so a pair of fresh engines is \$18.2–20.6M on those figures; no appraised value for a 12-to-15-year-old 737-800 was retrieved to set beside it (D6 U9). The only per-aircraft dollar figures in the research are the capital per aircraft in FTAI's 2025 vehicle, which range from \$11.9M (the seed portfolio, \$549M over 46 aircraft) to \$23.6M (\$2.1B over 89 aircraft under contract), each derived in the research from a different disclosure and none an appraisal; whether "total capital" includes reserves is unstated. The juxtaposition shows the order of magnitude and stops there. Limits of the example: dates and bases are mixed on both sides; the engine rents are a contested pair; the aircraft value is unknown. [IBA, Apr 2024; IBA, Jul 2025; Safe Fly Aviation, 2026; Part I worked example 1.7; derived in D7 §2.3; reconciliation R-028, R-056; D6 T5]

WHY THE ENGINES CARRY THE VALUE

Four facts from Part I combine. The engines are separable from the airframe and from each other at their module flanges, so they can be sold or leased on their own. Their value is dominated by maintenance condition, so a pair fresh from restoration with full life-limited parts is worth several million dollars more than the same pair run-out. No new CFM56 is being built for an airframe, so every serviceable -7B and -5B that will ever exist already exists, while the airframes they power are being replaced. And the airframe at 15 to 20 years has few other uses, which is why the 21-year-old 737-800 rents for "less than \$160,000" against \$228,000 at 12 years while a -7B rents for the same whether it hangs on a 12-year-old wing or a 21-year-old one. The consequence FTAI states in its own words is that it "owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices", and the consequence for its third-party vehicles is that the vehicles buy the aircraft and FTAI does the engine work (Part VIII §61). [Part I §3, §8; IBA, Jul 2025; FTAI Q2 2026 release, 29 Jul 2026; D6 §2.4]

The floor under an engine's value is what it fetches taken apart. A consultancy blog puts the price of a run-out CFM56 bought as teardown feedstock at \$0.8–1.2M and the used material it yields at \$1.6–2.2M, and the same blog's "green-time value" of \$2.8–3.4M for a -7B is a floor for engines with little life left; a full-life engine is worth substantially more. [Safe Fly Aviation, 2026 (consultancy blog)] So even at the end of an airframe's life, a pair of run-out -7Bs fetches \$1.6–2.4M as feedstock and yields \$3.2–4.4M of used material on those figures, before counting any life they still have. Section 16 takes up the trade in engines between those two states.

15.4 FREIGHTER CONVERSION AS AN END-OF-LIFE PATH

A passenger 737-800 at 18 to 25 years has one more use before part-out.

FREIGHTER CONVERSION, OR PASSENGER-TO-FREIGHTER (P2F)

Freighter conversion is the structural modification of a used passenger airframe into a cargo aircraft: a main-deck cargo door, a strengthened floor, a rigid barrier between the cockpit and the hold, and a cargo loading system. It extends the airframe's revenue life by 10 to 15 years at low cycles a year. Narrowbodies are converted at 21–22 years old on average; more than 260 Boeing 737-800 freighters were in service in mid-2025 with a pipeline exceeding 300. The conversion cost and the availability of conversion slots were not retrieved (D6 U10). [Cargo Facts, "Passenger-to-freighter conversions are taking place on increasingly older aircraft"; Acumen Aviation, 18 Dec 2025; Cargo Facts, "In-service 737NG freighter fleet surpasses 100"; D6 §2.4]

SUPPLEMENTAL TYPE CERTIFICATE (STC)

A Supplemental Type Certificate is a regulator's approval of a change to a certified aircraft design. A freighter conversion is performed under an STC owned by the conversion house, which is why conversion capacity belongs to a few STC holders. In Part VI §37 the same instrument is one of the approval bases for alternative parts. [D6 §2.4; D4 §2.2]

Conversion changes what the engines are asked to do. A freighter flies few cycles a year, so cargo operators want engines "customized for cargo": lower-cycle, lower-cost worksopes that burn remaining life rather than restore it. On 7 July 2026 FTAI and Aeronautical Engineers Inc. (AEI), a conversion house with "625+ aircraft modified using AEI's STCs", announced a collaboration on a 737-800 freighter solution built on such engines; Part VIII §67 carries it. [FTAI and AEI release, 7 Jul 2026] Two signals about the conversion market sit side by side in the research and are carried as a pair. The type-specialist data has more than 260 737-800 freighters in service and a pipeline above 300, and an earlier appraiser note had freighter values rising while passenger narrowbody rents receded. The same appraiser warned in September 2026 of an "oversupply of narrowbody conversions". [Acumen Aviation, Dec 2025; Cargo Facts; IBA via Aeroin, 2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-034] The primer records both and takes no view on which describes the market. What the conversion path does for this primer's subject is mechanical: it keeps -7B engines on wing past passenger retirement age, in a low-cycle duty that suits green-time modules, and it removes those airframes and engines from the teardown pool for a decade or more. Part III §24 carries that as a row of the translation table.

16. Green time and the trade in part-worn engines

Between a fresh engine worth its half-life value plus an adjustment and a run-out engine worth its parts lies a market in time. This section explains what green time is, how it is leased and traded, and how its value differs from the appraiser's half-life figure.

16.1 GREEN TIME, THE THREE CLOCKS, AND THE ENGINES THAT HAVE IT

Part I §8.2 boxed green time: the remaining usable life on an engine or module before its next mandatory shop visit, set by whichever runs out first. The research describes three clocks. The first is the life-limited part with the fewest cycles remaining, because the engine must come off the wing before any such part reaches its limit (Part I §4). The second is exhaust-gas temperature (EGT) margin, the gap between the temperature the engine runs at and its red line, which erodes as the engine wears and which a performance restoration recovers (Part I §5). The third is any hard-time inspection or airworthiness directive that requires the engine to be opened. [D3 §2.3; D2 §2.3; registry C-2] Green time is the quantity; a green-time engine, or a green-time lease, adds an intention. A green-time engine is "one that still has usable time but whose owner does not intend to pay for the next shop visit, either because the engine is near the end of its life or because the owner's fleet is leaving", sold or leased as-is to run down the remaining time and then torn down. [D3 §2.3]

USED SERVICEABLE MATERIAL (USM)

Used serviceable material is a part taken from a retired or dismantled aircraft or engine, inspected and if necessary repaired to the manufacturer's manual, and released on an airworthiness tag with its full history so that it can be installed again. A part with a gap in its paperwork is scrap whatever its condition. Used material is made by the original manufacturer, so it sits outside the alternative-parts debate, but it competes for the same slot in a shop visit's parts bill; its supply is a lagged function of retirements. Part V §31 carries the full treatment. [D3 §2.1; D2 §8]

RUN-OUT (EXHAUSTED) ENGINE

A run-out or exhausted engine is one with no usable time left before a shop visit: one of its clocks has reached zero. It is the only kind of engine that becomes teardown feedstock while engines with life are leased or sold whole, because a half-life -7B at \$5.7–6.4M (IBA, September 2025) is worth several times the \$1.6–2.2M of used material a teardown yields (consultancy blog, 2026). Part V §33 carries the full treatment. [D3 §2.2, §2.3; IBA, Sept 2025; Safe Fly Aviation, 2026]

16.2 THE GREEN-TIME LEASE

A green-time lease is a short fixed-term operating lease, from months to a few years, at a monthly rent, with the engine returned at or near run-out rather than restored. Because the lessee will not restore the engine, the rent is effectively the price of the time consumed, and the return conditions of §12.4 are inverted: the lessor expects the engine back with less life, not more. [D3 §2.3] The sequence has a sourced example. The engine trader APOC Aviation (APOC) leased a CFM56-5A to Condor for twelve months on a green-time lease, with the engine "expected to return to APOC's portfolio for teardown and part-out next year after the lease concludes".

[AviTrader, 28 Jul 2025; Aviation Business News, 2025]

The rents the research reached for green-time engines are the consultancy blog's, labelled as such: CFM56-7B \$42,000–48,000 a month, "about 34% above 2019" and up 15–20% "in recent months"; CFM56-5B \$38,000–44,000. [Safe Fly Aviation, 2026 (consultancy blog)] Whether those are green-time rents is the research's inference from the page's layout; the appraiser's \$100,000 spare-engine rent is the other side of the pair (§14.3, R-028). The appraiser's September 2025 release describes the -7B market as having "a distinct lack of availability", with market values and lease rates above long-term trend, which is the condition in which green-time engines are in demand: when shops are full, "operators often turn to short-term or green-time engine leasing". [IBA, Sept 2025; MTU AEROREPORT, undated] The engine maker sells the same thing: green-time leases are one element of GE's TrueChoice Transitions menu, which Part IV §28 carries. [D2 §2.3]

16.3 THE OWNER'S CHOICE, AND THE TRADE IN PART-WORN ENGINES

An engine lessor faces the same decision at every lease end, and a mid-life aircraft owner faces it for each engine when the airframe retires: restore the engine and lease it again at a full-life rent, lease out its remaining green time, or part it out now. [D6 §2.3] The trade in part-worn engines is the market in which those three options are priced against each other, and it is where the module as a unit of trade (Part I §8; Part V §36) comes from: a core with 80% of its life-limited-part life is worth far more than one with 20%, because the next owner's shop visit will or will not have to open it. [D3 §2.7]

WORKED EXAMPLE 2.7 — TWELVE MONTHS OF GREEN TIME: LEASE IT OUT OR TEAR IT DOWN NOW

Inputs: a CFM56-7B with twelve months of green time (assumed). Green-time rent \$45,000 a month, the midpoint of \$42,000–48,000 (sourced: Safe Fly Aviation, consultancy blog, 2026; the midpoint is the research's choice for illustration), with the appraiser's spare-engine rent of about \$100,000 a month (sourced: IBA, April 2024; a spare engine, not stated to be green-time) as the other side of the pair. Teardown yield \$1.6–2.2M of used material (sourced: Safe Fly Aviation, 2026).

lease twelve months, then tear down:
 rent at \$45,000: $12 \times 45,000 = \$540,000$ (at \$42,000–48,000: \$504,000–576,000)
 total: $540,000 + 1,600,000$ to $2,200,000 = \$2.14\text{M}$ to $\$2.74\text{M}$, over a year
 plus the sell-through period
 the same at the IBA rent: $12 \times 100,000 = \$1,200,000$; total $\$2.8\text{M}$ to $\$3.4\text{M}$
 tear down now: $\$1.6\text{M}$ to $\$2.2\text{M}$, sooner, with no rent

Leasing adds the rent but delays the teardown proceeds by a year and more, and the life-limited parts with twelve months of cycles left are worth slightly more in a teardown now than after another year of flying, since their remaining life is what a buyer pays for; the airfoils and cases are worth the same either way. Which option is chosen depends on the owner's cost of capital, the rent on offer, and whether an engine shop will pay more for the life-limited-part life now than a lessee will pay in rent.

On the blog's rent the lease adds a quarter to a third to the proceeds; on the appraiser's it adds more than half; the primer carries both and takes no view on which rent a green-time engine commands. Limits of the example: the twelve months are assumed; the two rents are a contested pair that may price different engines; the sell-through period for teardown inventory was not sourced (D3 U2); and the teardown yield is a consultancy figure with no appraiser counterpart. [D3 §2.3; Safe Fly Aviation, 2026; IBA, Apr 2024; reconciliation R-028]

16.4 HOW GREEN-TIME VALUE DIFFERS FROM HALF-LIFE VALUE

The appraiser's half-life market value and the consultancy blog's green-time value are not two estimates of one quantity. They are two quantities. Part I §8 gave the arithmetic that connects them: an engine's maintenance-adjusted value is its half-life value plus, for each clock, the actual remaining fraction less one half, times the cost to restore that clock. An engine with full-life life-limited parts is worth the half-life value plus half the full-life set; a run-out engine is worth the half-life value less half the set less half a restoration, which for an old engine "is approximately the teardown value". [D3 §2.4; Part I §8.1] A green-time engine sits between those points, and its value falls toward the run-out figure as its clocks run down.

EXHIBIT 2.7 — GREEN-TIME VALUE AGAINST HALF-LIFE VALUE: THE FIGURES THE RESEARCH REACHED, THE BASES THEY SIT ON, AND THE RUN-OUT VALUE PART I'S INPUTS IMPLY

Quantity	Figure	Basis	Source and level	Date
CFM56-7B27 / -7B24 market value	\$6.4M / \$5.7M	Half-life: every module midway between restorations, every LLP at half its life	IBA (appraiser)	Sept 2025
Full-life -7B27 on Part I's inputs	\$10.3M (\$9.1M on the alternative set and restoration prices)	Half-life plus half of (\$5.7M set + \$2.15M restoration)	Derived in Part I worked example 1.7	2025 prices
Run-out -7B27 on Part I's inputs	\$2.5M (\$3.7M on the alternative inputs)	Half-life less half of (\$5.7M + \$2.15M)	Derived in this Part	2025 prices
Run-out -7B24 on Part I's inputs	\$1.8M (\$3.0M on the alternative inputs)	Same, on the \$5.7M half-life value	Derived in this Part	2025 prices
CFM56-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", which the research notes is internally inconsistent with green time	Safe Fly Aviation (consultancy blog)	2026

Quantity	Figure	Basis	Source and level	Date
CFM56-5B "green-time value"	\$2.6–3.2M	Same label	Safe Fly Aviation	2026
Run-out CFM56 purchase price	\$0.8–1.2M	Exhausted engine bought for teardown	Safe Fly Aviation	2026
Used material from a run-out CFM56	\$1.6–2.2M; net \$0.4– 0.8M after the purchase	Teardown yield	Safe Fly Aviation	2026
LLP life as share of a used engine's value	60–80%; two -7Bs at 10,000 cycles since new differ by \$1.6M on LLP life alone	Used engine	Safe Fly Aviation	2026
Value direction	"Levelled off after approximately 18 months of increases"	-5B and -7B market values	IBA H1 2026	H1 2026
Value direction	Up "as much as 50% over the past two years"; rents up 15–20% "in recent months"	Not stated; green-time rents	ePlaneAI (trade press), date not captured; Safe Fly Aviation, 2026	2026

Source: IBA, Engine Values Release 2025B, Sept 2025; IBA, Engine and Lease Rate Update H1 2026; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); ePlaneAI, "CFM56 engine values highlight investor opportunities and risks", date not captured; Part I worked example 1.7 on IBA, MyAirTrade (June 2025), Aircraft Value News (Nov 2019) and Safe Fly Aviation inputs. Reconciliation R-029. Accessed 2026-10-03.

The derived rows use Part I's inputs and carry Part I's caveats: the life-limited-part set is \$5.7M (a market-data table, June 2025, scope not visible) or \$4.0M (a trade-press headline, November 2019), the restoration is \$2.15M or \$1.4M (two consultancy ranges), and the half-life reference is a market value standing in for a base value the research never reached. [Part I §8.3; reconciliation R-013, R-016, cluster K-02]

run-out -7B27 = $6.4M - 0.5 \times 5.7M - 0.5 \times 2.15M = 6.4 - 2.85 - 1.075 = \$2.5M$
 alternative inputs: $6.4M - 0.5 \times 4.0M - 0.5 \times 1.4M = 6.4 - 2.0 - 0.7 = \$3.7M$
 run-out -7B24: $5.7 - 2.85 - 1.075 = \$1.8M$; alternative: $5.7 - 2.7 = \$3.0M$

Read against the consultancy blog's figures, the derived run-out value of \$1.8–3.7M sits above the blog's \$0.8–1.2M purchase price for an exhausted engine and overlaps its \$2.8–3.4M "green-time value". The research does not reconcile these, and the primer does not. [reconciliation R-029]
 The bases differ in three ways. The appraiser's figure is a half-life market value and the blog's two figures are a green-time value and a feedstock price, so none of the three measures the same engine. The blog's label "(new condition)" on a green-time figure is internally inconsistent, as the

research noted. And the sources disagree on direction, the appraiser's "levelled off" against the blog's rents up 15–20% "in recent months" and a trade-press item's values up by half in two years. [IBA, H1 2026; Safe Fly Aviation, 2026; ePlaneAI, date not captured; D3 T3] What the arithmetic does establish is why the two sides can both be describing something real: the gap between a half-life -7B27 at \$6.4M and a run-out one at \$2.5M is about \$3.9M on Part I's inputs, and that gap is the restoration and the life-limited parts the appraiser's convention assumes are half consumed. A green-time engine is one whose owner has decided not to buy them back.

For FTAI the two ends of this trade are inputs to the same business. Engines it judges "unserviceable for leasing purposes" are "transferred to inventory ... with the intent to tear down the asset into salvageable components that would be manufactured into new modules and parts and sold", and serviceable used modules and parts are "sold through an exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool". [FTAI 10-K FY2025] Part VII §46 carries the module side and Part V §33 and §35 the partnership, which the research identifies from a partner's release as an exclusive Serviceable Engine Products agreement with the parts distributor AAR Corp (AAR), extended through 2030. [AAR release via PRNewswire, 2025 (title located; body not read); reconciliation B-9] The research also records FTAI on the other side of the trade, selling a 15-year-old A320-200 to APOC for disassembly in May 2026. [ePlaneAI, 2026]

17. Financing: securitizations, warehouse facilities, sale-leasebacks (worked tranching example)

A lessor's return in worked example 2.3 had a financing line of 4.55% of value, and the equity return of 13% depended on borrowing 70% of the price at 6.5%. Where that debt comes from, how it is layered, and what it costs the equity when rents fall is the subject of this section. The instruments are general to the industry, and the sourced template is the engine-lessor's: Willis Lease's rated securitizations. FTAI's own vehicles use the same instruments, and the facts specific to them belong to Part VIII; this section defines the instruments and points there.

17.1 THE VEHICLE THAT HOLDS THE ASSETS

Almost every form of asset-level debt in this industry starts with a company that exists only to own a defined set of aircraft or engines.

SPECIAL-PURPOSE VEHICLE (SPV)

A special-purpose vehicle is a bankruptcy-remote legal entity that exists only to own a defined pool of assets and the debt secured on them, insulated from the sponsor's other liabilities. Bankruptcy-remote means that if the sponsor fails, the SPV's assets are not caught up in the sponsor's insolvency, so a lender to the SPV looks only to those assets and their leases. It is the issuer in a securitization, the borrower under a warehouse facility, and the asset-owning layer beneath a partnership such as FTAI's Strategic Capital vehicles, which FTAI's filings describe as acquiring aircraft through "special purpose entities (SPVs) of the 2025 Partnership". Aircraft lessors are usually organised this way, with each aircraft or sub-pool in its own entity so that lenders and lessees deal with a clean owner. Part VIII §61 and §62 carry FTAI's vehicles. [FTAI 10-Q Q1 2026; definition assembled in the research, D6 §2.5, D7 §2.1]

The SPV is the reason a lessor can borrow against a pool of leased engines at a cost that reflects the engines and the airlines renting them rather than the lessor's own credit. Everything in this section is a way of lending to such a vehicle.

17.2 THE SECURITIZATION AND ITS PARTS**ASSET-BACKED SECURITIZATION (ABS)**

An asset-backed securitization is a transaction in which a lessor sells a portfolio of leased aircraft or engines to a bankruptcy-remote SPV, which issues notes to investors secured on the assets and on their lease cash flows. The notes are divided into tranches paid in order of seniority. The lessor usually keeps the equity, the residual claim after the notes, and acts as servicer of the leases for a fee. Rating agencies, KBRA (Kroll Bond Rating Agency) in the two deals this section uses, assign ratings to the notes after testing the structure against stressed assumptions for lease-rate declines, downtime, maintenance costs and residual values. [KBRA, 3 Jun 2025 and 10 Dec 2025; definition assembled in the research, D6 §2.5]

TRANCHE: SENIOR, MEZZANINE AND EQUITY

A tranche is one layer of the claims on a securitization, paid in order of seniority. The senior notes (Class or Series A) are paid first, carry the lowest coupon and the highest rating. The mezzanine notes (Class B, and sometimes C) are paid after the A notes and carry higher coupons and lower ratings. The equity, sometimes issued as E-certificates, is paid last and absorbs the first losses. Because every dollar of shortfall hits the equity first and every dollar of surplus reaches it last, tranching concentrates both the upside and the downside in the layer the lessor keeps. [definition assembled in the research, D6 §2.5; KBRA, 10 Dec 2025]

LOAN-TO-VALUE (LTV)

Loan-to-value is the note balance divided by the appraised value of the assets in the pool. The senior notes might be sized to 60–70% of appraised value and the senior and mezzanine notes together to 75–80%, leaving equity of 20–25%; those ranges are illustrative, because the two deals the research retrieved did not return their loan-to-value ratios (D6 U6). It is distinct from loan-to-cost, which divides debt by the total capital of a vehicle and which Part VIII §61 uses for FTAI's 2025 vehicle. [definition assembled in the research, D6 §2.5; D7 §2.3]

PAYMENT WATERFALL (ABS)

The payment waterfall is the ordered allocation of the pool's collections on each payment date: fees and expenses first, then interest on the A notes, then interest on the B notes, then scheduled principal on the A and B notes, then any C notes, with any surplus to the equity. Part VIII §63 uses the same word for a partnership's distribution order, which allocates cash among partners rather than among note classes; the primer says "payment waterfall" for the first and "distribution waterfall" for the second. [definition assembled in the research, D6 §2.5; registry C-17]

ANTICIPATED REPAYMENT DATE (ARD) AND LEGAL FINAL MATURITY

The anticipated repayment date is the date after which a securitization sweeps all excess cash to pay down principal faster and steps up the coupon on the notes, so that the deal is designed to be refinanced or called at that date. The legal final maturity is the much later last date by which the notes must be repaid. In Willis Lease's ninth engine securitization the two dates are December 2031 and December 2050, a gap of about nineteen years: the structure expects to be refinanced at the first date and treats the second as a backstop. [KBRA, 10 Dec 2025; Asset Securitization Report, Dec 2025]

The rating agencies are the gate. The notes carry ratings because an agency, KBRA in the two deals below, has modelled the pool's rents, the lessees' credit, the time engines spend off lease between lessees, the maintenance they will need and the value they will have at the end, under stresses, and found the senior notes able to pay through those stresses. [KBRA, 3 Jun 2025; KBRA, 10 Dec 2025] That is why the lease-rate and value movements of §14 and §15 matter beyond the lessor's own income statement: an appraiser's report that engine values have "levelled off" or that narrowbody rents fell 11–13% feeds straight into the stress tests that decide how much senior debt a pool can carry. [IBA, H1 2026; IBA, Jul 2025; D6 §5]

17.3 THE SOURCED TEMPLATE: WILLIS ENGINE STRUCTURED TRUST VIII AND IX

WILLIS ENGINE STRUCTURED TRUST (WEST)

The Willis Engine Structured Trust deals are Willis Lease's series of rated engine securitizations: WEST VIII, rated in June 2025, raised \$596M against 64 assets; WEST IX, rated in December 2025, raised \$392.9M against 49 assets. They are the sourced template in this primer for a senior-subordinate engine securitization, and Willis is their servicer and administrative agent, receiving 11.5% of aggregate net rents as servicer plus 2.0% as administrative agent each month and 3.0% of net proceeds on engine sales, according to its 10-K filings for 2019 and 2020. Part XI §87 reuses the deals in the peer comparison. [KBRA, 3 Jun 2025 and 10 Dec 2025; Asset Securitization Report, 2025 and Dec 2025; WLFC 10-K FY2019 and FY2020]

EXHIBIT 2.8 — WEST VIII AND WEST IX: THE STRUCTURE AS THE RATING AGENCY AND TRADE PRESS DESCRIBED IT, AND WHAT WAS NOT RETRIEVED

Item	WEST VIII	WEST IX	Source
Preliminary ratings announced	3 Jun 2025	10 Dec 2025	KBRA
Size	\$596M	\$392.9M	Asset Securitization Report
Collateral	64 assets; proceeds "to refinance the existing WEST IV transaction and acquire a portfolio"	"Financing leases on 49 aircraft assets" (the trade report's wording)	KBRA; Asset Securitization Report
Senior notes	Series A, rated A	Class A, rated A	KBRA
Subordinated notes	Series B, rated BBB (a grade below A)	Class B; "if a series of C notes is issued, it will receive payments subordinate to payments to the series A and B"	KBRA
Repayment structure	Senior-subordinate	"A senior-subordinate repayment structure"	KBRA
Anticipated repayment date	Not returned	December 2031	KBRA
Legal final maturity	Not returned	December 2050	KBRA
Sponsor and servicer	Willis Lease; "the ninth aviation ABS sponsored and serviced by WLFC"	Willis Lease; the tenth	KBRA

Item	WEST VIII	WEST IX	Source
Servicer and administrative-agent fees	11.5% + 2.0% of aggregate net rents, monthly; 3.0% of net engine-sale proceeds	Same source; whether unchanged for WEST IX not retrieved	WLFC 10-K FY2019, FY2020
Note balance per asset (derived)	$\$596\text{M} \div 64 = \9.3M	$\$392.9\text{M} \div 49 = \8.0M	Derived in this Part; a note balance, not a value
Coupons, loan-to-value, equity size	Not returned (D6 U6)	Not returned (D6 U6)	KBRA pre-sale reports not read

Source: KBRA, "Assigns Preliminary Ratings to Willis Engine Structured Trust VIII", BusinessWire, 3 Jun 2025; KBRA, "Assigns Preliminary Ratings to Willis Engine Structured Trust IX", BusinessWire, 10 Dec 2025; Asset Securitization Report (American Banker), "Willis Engine Structured Trust raises \$596 million in aircraft ABS", 2025, and "Willis Engine floats \$392.9 million in aircraft-related ABS", Dec 2025; Willis Lease Finance Corp., Form 10-K FY2020 and FY2019 (WEST fees). Accessed 2026-10-03.

17.4 THE TRANCHING WORKED THROUGH

The research retrieved the sizes, the ratings and the dates of the two deals but not their coupons, their loan-to-value ratios or the size of their equity (D6 U6). The example below therefore takes WEST IX's note total as its one sourced size and applies the illustrative structure the research set out, with every other input labelled assumed. It is arithmetic on a real deal's size, not a description of the deal.

WORKED EXAMPLE 2.8 — TRANCHING A \$392.9M ENGINE SECURITIZATION ON WEST IX'S SIZE, AND WHAT A FALL IN RENTS DOES TO EACH LAYER

Inputs. Total notes \$392.9M on 49 assets (sourced: KBRA and Asset Securitization Report, December 2025). Anticipated repayment date December 2031, legal final December 2050 (sourced: KBRA). Servicer and administrative-agent fees 13.5% of net rents (sourced rate: WLFC 10-K FY2019 and FY2020; applied here to gross rent as an approximation, since the definition of "net rents" was not retrieved). Everything below is assumed for illustration: the senior and subordinated notes together are 75% of appraised value; the Class A notes alone are 65%; the A coupon is 5.5% and the B coupon 7.5%; the pool earns a lease rate factor of 1.0% a month; other expenses (insurance, remarketing, downtime) are \$3.0M a year; scheduled principal is 4% of the notes a year.

implied appraised value: $392.9 \div 0.75 = \$523.9\text{M}$ (per asset: $523.9 \div 49 = \$10.7\text{M}$)
 Class A: $0.65 \times 523.9 = \$340.5\text{M}$ (65% LTV)
 Class B: $392.9 - 340.5 = \$52.4\text{M}$ (A+B: 75% LTV)
 equity: $523.9 - 392.9 = \$131.0\text{M}$ (25%)
 annual interest: A $0.055 \times 340.5 = 18.7$; B $0.075 \times 52.4 = 3.9$; total $\$22.6\text{M}$
 annual rent at 1.0% a month: $0.01 \times 523.9 \times 12 = \62.9M
 servicer and agent fees: $0.135 \times 62.9 = \$8.5\text{M}$

The waterfall in a normal year:

rent 62.9 – fees 8.5 – other expenses 3.0 = 51.4 available for the notes
 – A interest 18.7 – B interest 3.9 = 28.8 after interest
 – scheduled principal 15.7 = 13.1 to the equity
 equity cash yield: $13.1 \div 131.0 = 10.0\%$
 coverage of interest plus scheduled principal: $51.4 \div (22.6 + 15.7) = 1.34\times$

Now stress the rent as a rating agency would. If lease rates fall 15%, rent is \$53.4M and the fees fall with it to \$7.2M; the notes' interest and scheduled principal do not move.

at -15%: $53.4 - 7.2 - 3.0 - 22.6 - 15.7 = \4.9M to the equity (3.7%);
 coverage $43.2 \div 38.3 = 1.13\times$
 at -25%: $47.2 - 6.4 - 3.0 - 22.6 - 15.7 = -\0.5M (nothing to the equity);
 coverage $37.8 \div 38.3 = 0.99\times$

A 15% fall in rent cuts the equity's cash by nearly two-thirds, from \$13.1M to \$4.9M, while the A and B notes are paid in full. A 25% fall leaves nothing for the equity and a small shortfall against scheduled principal, which is the point at which a real structure's reserve accounts and cash-trap triggers would operate; they are not modelled here. That asymmetry is the tranching doing its job: the equity concentrates the downside so that the senior notes can carry an A rating, and it concentrates the upside, which is why the lessor keeps it. After the anticipated repayment date in December 2031 the surplus that reached the equity in a normal year would instead be swept to principal and the coupons would step up, so the deal is built to be refinanced then. For scale, WEST VIII's \$596M over 64 assets is \$9.3M of notes per asset against WEST IX's \$8.0M, both derived; neither is a value, and the pools include engine types and aircraft the research did not itemise, so neither should be set against the IBA half-life market values of \$5.7–6.4M for a -7B. Limits of the example: the 75% and 65% loan-to-value ratios, both coupons, the lease rate factor, other expenses and the principal schedule are all assumed; the fee rate is from filings five and six years old; the actual coupons, ratios and equity of WEST IX were not retrieved (D6 U6). [KBRA, 10 Dec 2025; Asset Securitization Report, Dec 2025 and 2025; WLFC 10-K FY2019 and FY2020; illustrative structure from D6 §2.5]

17.5 THE WAREHOUSE FACILITY

A securitization needs a pool large and seasoned enough to rate. The instrument that assembles the pool is a bank line.

WAREHOUSE FACILITY (WAREHOUSE LINE)

A warehouse facility is a revolving bank loan to an SPV, secured on whatever assets the SPV buys during an availability period, with an advance rate and a borrowing base, and with a term after which the balance must be repaid or refinanced, commonly by issuing a securitization. Assets are accumulated in it until there are enough to securitize, which is why it is called a warehouse. A lessor uses it to fund acquisitions one aircraft at a time without going to the bond market for each. FTAI's second Strategic Capital vehicle closed a \$2.0B warehouse facility from 13 lenders on 14 August 2026; Part VIII §62 carries the vehicle. [FTAI release, 17 Aug 2026; definition assembled in the research, D6 §2.5, D7 §2.2]

ADVANCE RATE AND BORROWING BASE

The advance rate is the percentage of each asset's value that the banks will lend against it under a warehouse facility. The borrowing base is the total credit available at any moment, recomputed as assets are added, sold or revalued: the sum over the assets of value times advance rate. If values fall, the borrowing base falls, and the borrower must repay or add collateral. [definition assembled in the research, D6 §2.5]

ACCORDION

An accordion is a pre-agreed option to enlarge a credit facility if lenders agree to fund more. The 2026 vehicle's \$2.0B warehouse carries a \$1.0B accordion, so up to \$3.0B. FTAI's statement that its Strategic Capital vehicles "have now raised \$5.5 billion of warehouse financing in less than two years" matches the disclosed commitments of \$2.5B (February 2025) and \$2.0B (August 2026), which sum to \$4.5B, only if the undrawn accordion is counted; the release does not say how the figure is built. Part VIII §62 carries the full treatment. [FTAI release, 17 Aug 2026; FTAI release, 26 Feb 2025; reconciliation R-055]

WORKED EXAMPLE 2.9 — WHAT A WAREHOUSE LINE BUYS: ADVANCE RATES, THE ACCORDION, AND A VEHICLE'S LOAN-TO-COST

Inputs: a \$2.0B warehouse facility with a \$1.0B accordion (sourced: FTAI release, 17 Aug 2026). Advance rates of 70% and 60% (assumed; the facility's rate, tenor and covenants were not retrieved, D7 U11). For the first vehicle, about \$6.0B of total capital on \$2.0B of equity commitments (sourced: FTAI releases, 27 Oct 2025 and 17 Aug 2026) and a \$2.5B asset-level debt commitment (sourced: FTAI release, 26 Feb 2025).

assets a \$2.0B line funds at a 70% advance rate: $2.0 \div 0.70 = \$2.86\text{B}$,
with \$0.86B of equity
with the \$1.0B accordion drawn: $3.0 \div 0.70 = \$4.29\text{B}$, with \$1.29B of
equity
at a 60% advance rate: $2.0 \div 0.60 = \$3.33\text{B}$ (\$1.33B equity); $3.0 \div 0.60 =$
 $\$5.00\text{B}$ (\$2.00B equity)
first vehicle, loan-to-cost: debt = $6.0 - 2.0 = \$4.0\text{B}$; $4.0 \div 6.0 = 67\%$,
or \$2 of debt per \$1 of equity
if only the \$2.5B commitment were drawn: total 4.5; $2.5 \div 4.5 = 56\%$

The advance rate sets how far a dollar of equity stretches: at 70% each equity dollar buys \$3.33 of aircraft, at 60% \$2.50. The accordion adds buying power only if the lenders agree to fund it, which is why the primer carries "\$5.5 billion" and "\$4.5 billion" side by side rather than choosing. The first vehicle's 67% is a loan-to-cost figure derived in the research from two disclosed totals; it is not a loan-to-value, because no appraised value of the vehicle's aircraft was disclosed, and "total capital" may include reserves for engine exchanges. Limits of the example: the advance rates are assumed; the debt actually drawn by the first vehicle was not disclosed, so 56% and 67% bracket it; Part VIII §61 carries the vehicles' leverage in full. [FTAI releases, 26 Feb 2025, 27 Oct 2025 and 17 Aug 2026; derived in D7 §2.3; reconciliation R-055, U-D7-11]

EXHIBIT 2.9 — THE WAREHOUSE FACILITY AND THE SECURITIZATION, COMPARED AS INSTRUMENTS

Feature	Warehouse facility	Asset-backed securitization	Source
Lender	A bank or syndicate of banks (13 in the 2026 vehicle's facility)	Note investors, in rated tranches	FTAI release, 17 Aug 2026; KBRA
Borrower or issuer	An SPV	An SPV	D6 §2.5
Security	Whatever assets the SPV buys during the availability period	A fixed, identified pool of leased assets and their lease cash flows	D6 §2.5
Sizing	Advance rate on each asset; borrowing base recomputed as the pool changes	Loan-to-value on the appraised pool, by tranche	D6 §2.5
Pricing	Floating bank rate, not retrieved for FTAI's vehicles (D7 U11)	Fixed coupons by tranche, lowest on the senior notes; not retrieved for WEST (D6 U6)	D6 U6; D7 U11

Feature	Warehouse facility	Asset-backed securitization	Source
Term	Availability period, then repay or refinance	Anticipated repayment date (WEST IX: Dec 2031), legal final (Dec 2050)	KBRA, 10 Dec 2025
Flexibility	Revolving: draw as assets are bought, repay as they are sold; accordion to enlarge	Fixed at issue; refinanced at the anticipated repayment date	FTAI release, 17 Aug 2026; D6 §2.5
Role in the sequence	Accumulate and season the pool	Take out the warehouse; release capacity for the next pool	D6 §2.5
Who keeps the equity	The vehicle's equity holders	The sponsor, typically, as E-certificates	D6 §2.5
Example in the research	FTAI 2026 vehicle: \$2.0B, 13 lenders, \$1.0B accordion, closed 14 Aug 2026; first vehicle: \$2.5B asset-level debt commitment, 26 Feb 2025	WEST VIII (\$596M, 64 assets, 2025); WEST IX (\$392.9M, 49 assets, Dec 2025)	FTAI releases; KBRA; Asset Securitization Report

Source: FTAI releases of 26 Feb 2025 and 17 Aug 2026; KBRA, 3 Jun 2025 and 10 Dec 2025; Asset Securitization Report, 2025 and Dec 2025; instrument descriptions assembled in D6 §2.5 and D7 §2.2. Accessed 2026-10-03.

17.6 HOW A LESSOR USES THEM TOGETHER, AND WHAT THAT HAS TO DO WITH FTAI

The sequence a lessor follows is the same whether the lessor is Willis Lease or a vehicle managed by FTAI. Buy assets on the warehouse. Season the leases, meaning let them run long enough to show the collections a rating agency can model. Securitize the pool into a rated deal, which repays the warehouse and releases its capacity for the next pool. Keep the equity and the servicing fee. [D6 §2.5] Willis has done this ten times; the WEST VIII proceeds were partly used to refinance WEST IV, which is the same sequence one generation on. [KBRA, 3 Jun 2025]

SERVICER AND SERVICING FEE

The servicer is the party contracted to manage the assets of a vehicle it does not wholly own: leasing and remarketing them, collecting rent and reserves, overseeing their maintenance condition, and selling them, for a fee. Willis is the servicer of the WEST deals at 11.5% of net rents plus 2.0% as administrative agent and 3.0% of sale proceeds; FTAI is the Servicer of its Strategic Capital vehicles, "receiving customary, market-based compensation", which amounted to \$10.2M in 2025 and \$12.85M in the first half of 2026; the rate and base are undisclosed. Part VIII §61 and §63 carry the full treatment. [WLFC 10-K FY2019 and FY2020; FTAI 10-K FY2025; FTAI 10-Q Q2 2026; D7 §2.5]

FTAI's third-party capital programme is built from the instruments this section has defined, with a partnership layered on top. Three terms are needed to read it, and Part VIII owns all three.

STRATEGIC CAPITAL INITIATIVE (SCI)

The Strategic Capital Initiative is FTAI's asset-management arm, launched on 30 December 2024, which raises third-party capital into closed-end vehicles that buy on-lease, mid-life 737NG and A320ceo aircraft; FTAI manages the vehicles as Servicer, co-invests as a limited partner, and supplies the engine work on their fleets through its Maintenance, Repair and Exchange business. It is not a reportable segment. Part VIII §61 carries the full treatment. [FTAI release, 30 Dec 2024; FTAI 10-K FY2025; D7 §2.1]

THE 2025 PARTNERSHIP AND THE 2026 SPV

The 2025 Partnership is the first Strategic Capital vehicle: a closed-end partnership over asset-owning SPVs that closed at a \$2.0B equity hard cap in October 2025 (against a \$1.5B target), carries a \$2.5B asset-level debt commitment made in February 2025, and had committed about \$6.0B of total capital across more than 300 aircraft by August 2026, when FTAI described it as in its harvest phase; FTAI holds 19% (the FY2025 10-K and 2026 10-Qs) or 20% (the Q3 2025 10-Q) as a limited partner. The 2026 SPV is the second vehicle, which closed the \$2.0B warehouse facility above on 14 August 2026 and made its first disclosed purchase in the WestJet transaction of 28 September 2026; its equity size and FTAI's stake were undisclosed at the research date. Both followed the warehouse-then-securitize shape described above. Part VIII §62 carries the full treatment. [FTAI releases, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026 and 28 Sep 2026; FTAI 10-K FY2025; FTAI 10-Q Q3 2025; reconciliation R-048]

SEED ASSETS

Seed Assets is FTAI's term for the on-lease aircraft it sold from its own book to start the 2025 Partnership: 46 aircraft at "an estimated net purchase price of \$549 million" at launch. The sales are why FTAI's leasing equipment at cost fell from \$2,963.5M to \$2,057.6M during 2025 and why its lease income roughly halved year on year in the first half of 2026. Part VIII §60 carries the full treatment. [FTAI release, 30 Dec 2024; FTAI 10-K FY2025; FTAI 10-Q Q1 2026]

The vehicles carry two layers of debt in the sense this section has defined them: asset-level debt borrowed by the SPVs and secured on the aircraft and their leases, with no recourse to FTAI, and a warehouse facility within that layer for the second vehicle. [FTAI release, 26 Feb 2025; FTAI release, 17 Aug 2026; D7 §2.2] The lessor's incentive to sell assets into such vehicles is to recycle equity: it books a gain on sale, keeps a servicing stream and, in FTAI's case, keeps the engine maintenance work on the fleet it sold. [D6 §2.5; D7 §2.5] The equity it keeps is a 19% or 20% limited-partner stake rather than the whole E-certificate layer a securitization sponsor usually holds, which is one of the differences Part VIII §65 sets beside other platforms. The first vehicle has since used the securitization market itself. On 22 May 2026 FTAI priced FTAI MRE 2026-1, which it called

the Strategic Capital vehicle's inaugural asset-backed securitization: \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23 airlines, in two classes, Series A expected to be rated Asf by Fitch and A(sf) by KBRA and Series B expected BBB+sf by Fitch, with closing expected on 4 June 2026 and both classes described as significantly oversubscribed. The same release put the vehicle's fleet at 292 aircraft. So the deal is a take-out within the vehicle's asset-level layer, of the kind §17.6 describes, not a borrowing by FTAI itself. Its coupons and loan-to-value were not in the research (D7 U12, now partly resolved). [FTAI release, "FTAI Aviation Prices Inaugural Asset-Backed Securitization", 22 May 2026, via Finviz and AviTrader 26 May 2026]

17.7 SALE-LEASEBACKS

The third instrument is not a loan at all from the lessor's side, but it is the airline's way of financing an aircraft and the lessor's way of acquiring one.

SALE-LEASEBACK

A sale-leaseback is a transaction in which an airline sells aircraft it owns to a lessor and leases them straight back, converting owned assets into rent obligations and releasing cash to the airline. For the lessor it is the purest form of lease-attached acquisition: the aircraft arrives with a lease to a known operator already on it, which is what lenders and securitization investors underwrite. FTAI's WestJet transaction of 28 September 2026, in which the 2026 SPV acquired 17 Boeing 737-700s "on lease to WestJet in a sale-leaseback" while FTAI itself acquired 10 off-lease aircraft for their engines and modules, is the example in the research; the price and lease terms were not disclosed. Part VIII §67 carries it. [FTAI release, 28 Sep 2026; definition assembled in the research, D6 §2.4]

The sale-leaseback closes the loop this Part opened. The airline that sells its aircraft and rents it back is now a lessee under an operating lease (§12), paying rent and either reserves or end-of-lease compensation (§13), against return conditions that decide who pays for the engine life it consumes. The lessor that bought it funds the purchase on a warehouse line and later a securitization (§17), whose ratings rest on the appraisers' values (§12.7) and whose equity bears the swings in rent (§14, §15). And at the end of the lease the lessor faces the choice of §16: restore and re-lease, lease the green time, or take the engines out and sell them as modules, which is where Part VII begins.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-028	CFM56-7B rent: ~\$100,000 a month (IBA, Apr 2024, appraiser) against \$42,000–48,000 (Safe Fly Aviation, 2026, consultancy blog; may be green-time)	§12.1 box; §14.3, Exhibit 2.3, worked example 2.5; §15.3, worked example 2.6; §16.2, worked example 2.7
R-029	-7B value: half-life market value \$5.7M/\$6.4M and "levelled off" (IBA) against green-time \$2.8–3.4M "(new condition)", run-out \$0.8–1.2M, rents up 15–20% (Safe Fly) and values +50% in two years (ePlaneAI)	§14.3, Exhibit 2.3; §16.4, Exhibit 2.7
R-030	Lease rate factor: 0.70% (D3, green-time rent on half-life value); 0.87–1.1% (FTAI, derived on book); ~1.0% (Willis, derived); 0.9% (D7 assumed); 1.56–1.75% (IBA rent on IBA value, derived here); no stated figure	§12.1; §14.3, Exhibit 2.3, worked example 2.5; gross leasing yield box
R-031	Mid-life rents: 12-yr 737-800 ~\$228,000 (–11%), A320-200 ~\$220,000 (–13%) and a Sept 2026 oversupply warning (IBA) against rates up in 2025 and "expected to remain strong" (Air Lease 10-K FY2025), 10-yr 737-800 rents +44% vs Jan 2022 (Cirium Ascend) and extension rents >40% above prior (IBA, Nov 2023)	§15.1, Exhibit 2.6
R-032	Willis Lease: utilisation 86.0% at 30 Sep 2025 against 84.9% average for 2025; portfolio \$2,888.5M (Sep) against \$2,801.7M equipment (Dec); quarterly against annual rent and reserve revenue	§14.4, Exhibit 2.4
R-033	AerCap engines: 470 owned (20-F, 31 Dec 2025) against >1,200 including Shannon Engine Support (6-K, 30 Sep 2025)	§14.5, Exhibit 2.5
R-034	Freighter conversion: >260 737-800Fs in service, >300 pipeline (Acumen, Cargo Facts) and freighter values rising (IBA, 2025) against "oversupply of narrowbody conversions" (IBA, 6 Sep 2026)	§15.4
R-013	Restoration cost: D6's \$4.5M assumption against the sourced ranges \$1.2–1.6M and \$1.8–2.5M (Safe Fly, two pages)	§13.3, worked example 2.2
R-016 / R-017	LLP set behind a reserve or a maintenance adjustment: \$5.7M (MyAirTrade, Jun 2025) against ~\$4M (Aircraft Value News, Nov 2019)	§13.3; §16.4, Exhibit 2.7

Id	The two (or more) figures	Where in this Part
R-019	Turnaround: 90–120 days (trade press); ~90 sustaining (GE); 75–90 for a restoration, 120–150 heavy (Safe Fly); slot wait +2–3 months (Bain)	§14.2, worked example 2.4
R-008	Restoration interval: 10,000–15,000 cycles (Leeham) at 1,000–1,500 cycles a year, ~10 years implied (D9), ~8 years (Aviation Week), 6 years (D7 assumption)	§14.2, worked example 2.4
R-055	"\$5.5 billion of warehouse financing" (FTAI release, 17 Aug 2026) against \$2.5B + \$2.0B = \$4.5B disclosed; matches only with the \$1.0B accordion	§17.5 accordion box, worked example 2.9
R-056	Capital per aircraft in the 2025 vehicle: \$11.9M (seed), \$13.9M, \$14.7M, \$20M, \$23.6M on different bases; no appraised aircraft value retrieved	§15.3, worked example 2.6
R-048	FTAI's stake in the 2025 Partnership: 19% (FY2025 10-K, 2026 10-Qs) against 20% (Q3 2025 10-Q)	§17.6 box
D6 T5	Engines ≈88% of a 737-800's rent mixes a 2024 engine rent with a 2025 aircraft rent; on the Safe Fly rent the share is 37–42%	§15.3, worked example 2.6
D3 T3	Safe Fly's "green-time value (new condition)" label is internally inconsistent	§14.3, Exhibit 2.3; §16.4, Exhibit 2.7
Registry C-3	Utilisation: Willis by book value on lease (84.9%) against FTAI by equity-value-weighted days on lease excluding airframes (~77%); not comparable without the definitions	§14.4

Unknowns carried in this Part

- **U-D6-05 (D6) — U5 — Lease-rate factors.** No stated LRF for FTAI or WLFC; WLFC engine count, average lease term and FY2025 totals. Resolve: WLFC FY2025 10-K ([S13]) Item 1 and Item 7.
- **U-D6-06 (D6) — U6 — ABS coupons and LTVs.** WEST VIII and IX note coupons, initial LTVs, and E-certificate sizes. Resolve: KBRA pre-sale reports ([S16],[S17]) and WLFC 8-Ks.
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note.

- **U-D6-08 (D6) — U8 — Depreciation policy.** Useful lives and residual assumptions for FTAI's aircraft and engines. Resolve: FY2025 10-K summary of significant accounting policies.
- **U-D6-09 (D6) — U9 — Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values).
- **U-D6-10 (D6) — U10 — Conversion cost and slot availability** for 737-800 P2F (AEI, Boeing BCF, IAI). Resolve: conversion-house releases, trade press.
- **U-D6-14 (D6) — U14 — Current engine-lessor scale.** Portfolio size and value for GE Engine Leasing, SMBC Aero Engine Lease, ELF, AerCap Engines, RRPf, MTU Maintenance Lease Services. Resolve: company websites and parent filings (D11 for the listed ones).
- **U-D6-15 (D6) — U15 — Average lease terms over time** (new-lease term at signing versus remaining term) and the mix of short- versus long-term engine leases in FTAI's book. Resolve: 10-K MD&A by year.
- **U-D1-09 (D1) — U9** Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium). Resolve with an appraiser publication; this session's searches returned definitions, not values.
- **U-D3-02 (D3) — U2 — Total sell-through period** for teardown inventory (months to recover, say, 80% of proceeds). AerSale's 10-K inventory-turnover disclosure or a trade-press teardown case study would answer it.
- **U-D3-04 (D3) — U4 — Any mba Aviation (or Ascend by Cirium) CFM56-5B/-7B value** or lease-rate figure. None returned.
- **U-D3-05 (D3) — U5 — A current appraiser half-life value** for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).
- **U-D4-01 (D4) — U1 — Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks.
- **U-D7-11 (D7) — U11 — Debt terms.** Rate, advance rate, tenor and covenants of the \$2.5B facility and the \$2.0B warehouse; how \$5.5B is built; whether the 2025 facility was upsized.
- **U-D7-12 (D7) — U12 — The May 2026 ABS.** Whether "FTAI Aviation Prices Inaugural Asset-Backed Securitization" [S26] is a 2025-SPV take-out or FTAI's own balance-sheet deal; size, tranches, pricing.

- **U-D11-01 (D11) — U1 — Willis Lease lease rate factor and CFM56/LEAP portfolio split.** Searched the FY2025 10-K [S1]. Not returned. Resolve: read the 10-K's "Lease Portfolio" and "Market" sections directly.
- **U-D11-03 (D11) — U3 — Willis Lease maintenance services and spare-parts revenue lines for FY2025.** The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1].

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